

Japan IT Hardware & Electronics

Murata: MLCC Deep Dive - From ceramic to gold and the AI Data Center catalyst (Part 1)



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For years, Multi-layered ceramic capacitors (MLCCs) were volume based commoditized products that exist in consumer electronic devices. The AI Data Center (DC) demand for these electronic components has led to tight supply in high-end and edge MLCCs resulting in price hikes and high Book-to-Bill ratios. Our proprietary MLCC model is available upon request. In this Deepdive, we would look into the microscopic world of electronic components within the circuit board.

An electronic circuit board consists of active and passive components to regulate electrical power. Electrical power in a circuit is unstable from both the source and load. Active components detect the load changes and consumes additional power to correct the course. Passive components respond instantaneously through storing and releasing additional power supply. Capacitors are a core passive component aimed at regulating voltage, located across the circuit board and beside critical components. MLCCs are a sub-segment of capacitors where its dense dielectric properties of ceramic make them optimal for high performing devices. Automotive and computer industries require high-end MLCC capable of withstanding high temperatures, while communication and home electronic devices are low-end volume based.

Strict data center specifications has asymmetrically benefited high-end/cutting-edge MLCCs. MLCCs in the circuit boards of AI servers need to withstand high changes in power amidst tight spaces, ensuring reliability when operating in high temperature environment. The proximity of MLCCs surrounding and beneath GPUs make space as the main constraint, hence favoring the ceramic properties of MLCCs. As AI Data Center power grows by 19% CAGR till 2030E, we expect cutting-edge MLCCs demand to grow by 33% CAGR to USD27.7Bn by 2032E.

Difficulty in producing cutting-edge MLCCs would see supply tightness at least till 2030E. Producing cutting-edge MLCC requires miniaturization, material and manufacturing specialty. These specialties are complex and only few companies, Murata and Samsung Electro-Mechanics (SEMCO) - not covered - could produce them effectively at high yields. Capacity is estimated to grow at 50% CAGR attempting to catch up to demand growth at 30% CAGR through 2031E. As such, supply tightness will not be resolved at least in the next 5 years, resulting in potential further price uplifts.

Murata is the key beneficiary from AI Data Center demand. Murata's technological edge in miniaturization and ceramic manufacturing enables it to produce high-end and cutting-edge MLCCs (47-100 microfarad with the standard size of 0402-0603) effectively at lower costs. We estimate DC takes up almost 20% of Murata's Capacitor revenue while Autos take up almost 35%. SEMCO is weaker on high-end and even weaker on cutting-edge technology with a strategy to expand into both automotive and DC MLCCs. As such, we expect Murata to grab share in AIDC MLCC from 47% in 2025 to 76% in 2031E. Murata's MLCC mix shift and higher ASP from cutting-edge MLCC would drive total capacitor revenue up by +23% CAGR in the next 5 years.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	18 Sep 2026 Closing Price	Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
						Cur	2025A	2026E	2027E	2025A	2026E	2027E
6981.JP (Murata)	O	JPY	7,763.00	10,000	145.4%	JPY	127.66	205.12	292.61	60.8	37.8	26.5
JPL			2,660.83									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

MLCCs were previously unattractive investment areas due to global competition reducing profit margins for MLCC manufacturers. Historical valuation movements show cyclicity that follows the supply tightness of MLCCs lasting 1-2 years on average- but we think the current AI DC tightness will be longer at ~5 years. Murata's technological advantage is clearest in times of supply tightness from technological evolution. The AI data center and automotive upcycle today reveals where the gold is- in high end MLCC which Murata has the miniaturization technology and potential capacity advantage to capture market share and profit. **We rate Murata Outperform, TP JPY10,000**

VALUATION COMPS TABLE

EXHIBIT 1: Coverage Comps

Company	Ticker	Market Cap (US\$m)	CCY	Price	2025 P/S	2026E P/S	2025 P/E	2026E P/E	2026E revenue growth %	2026E operating margin %	2026E P/B
Japan IT Hardware & Electronics Coverage											
Hitachi Ltd.	6501 JP	159,786	JPY	5,524	2.4x	2.1x	31.3x	23.9x	15.1%	13.0%	3.3x
Mitsubishi Electric Corp	6503 JP	69,083	JPY	5,126	1.8x	1.6x	26.1x	17.7x	13.1%	10.5%	2.0x
Mitsubishi Heavy	7011 JP	83,459	JPY	3,879	2.6x	2.3x	11.7x	7.5x	13.4%	11.0%	3.4x
Kawasaki Heavy	7012 JP	13,448	JPY	2,405	0.9x	0.8x	18.6x	17.8x	11.8%	7.1%	1.9x
Murata Manufacturing	6981 JP	97,186	JPY	7,763	7.7x	6.5x	60.8x	37.8x	19.4%	21.7%	4.6x
NEC Corp	6701 JP	41,867	JPY	4,812	1.8x	1.7x	23.7x	19.6x	6.9%	12.0%	2.5x
Panasonic Holdings	6752 JP	67,114	JPY	4,287	1.2x	1.2x	52.8x	21.0x	1.3%	8.1%	1.8x
Median		69,083		4,812	1.8x	1.7x	26.1x	19.6x	13.1%	11.0%	2.5x
Japan IT Hardware Comps											
Daikin Industries Ltd.	6367 JP	38,770	JPY	20,740	1.2x	1.1x	20.8x	18.1x	4.6%	8.4%	1.9x
Canon Inc	7751 JP	38,303	JPY	4,503	1.3x	1.2x	11.9x	11.4x	2.2%	9.7%	1.1x
Fujitsu Ltd.	6702 JP	44,149	JPY	3,979	2.0x	1.9x	21.9x	19.0x	3.9%	11.7%	3.4x
Nikon Corp	7731 JP	3,715	JPY	1,746	0.8x	0.8x	52.5x	27.0x	6.0%	1.4%	1.0x
Median		38,537		4,241	1.2x	1.2x	21.3x	18.6x	4.2%	9.0%	1.5x
Global Defense Comps											
General Electric	GE US	331,211	USD	319	6.5x	5.9x	40.4x	35.2x	8.6%	21.9%	18.8x
RTX Corp	RTX US	257,408	USD	191	2.7x	2.5x	26.4x	24.2x	6.9%	13.6%	3.9x
Lockheed Martin Corp	LMT US	120,551	USD	522	1.5x	1.4x	17.1x	15.9x	5.5%	12.0%	13.7x
Hanwha Aerospace	012450 KS	38,823	KRW	1,018,000	1.6x	1.5x	22.4x	17.5x	8.4%	16.7%	4.4x
IHI Corp	7013 JP	19,698	JPY	2,853	1.7x	1.6x	16.7x	18.3x	5.9%	13.9%	4.3x
Subaru Corp	7270 JP	12,043	JPY	2,633	0.4x	0.4x	11.7x	9.4x	2.7%	3.3%	0.7x
Median		79,687		1,577	1.6x	1.5x	19.7x	17.9x	6.4%	13.7%	4.4x
Japan Electronics Comps											
SEMO	009150 KS	81,872	KRW	1,481,000	7.7x	5.8x	72.9x	34.4x	29.7%	21.5%	10.5x
Fujikura Ltd.	5803 JP	56,063	JPY	4,952	5.2x	4.7x	25.7x	23.3x	11.2%	24.6%	13.5x
TDK Corp	6762 JP	35,270	JPY	2,845	1.9x	1.7x	20.7x	17.7x	7.7%	11.7%	2.3x
Kyocera Corp	6971 JP	32,916	JPY	3,637	2.5x	2.4x	30.6x	27.6x	4.3%	7.3%	1.4x
Nidec Corp	6594 JP	21,722	JPY	2,856	1.3x	1.2x	17.7x	15.4x	4.3%	9.1%	1.9x
Taiyo Yuden	6976 JP	8,225	JPY	9,375	3.1x	2.7x	40.0x	22.6x	14.5%	10.4%	3.3x
Median		34,093		4,295	2.8x	2.6x	28.1x	23.0x	9.5%	11.1%	2.8x
Global IT Hardware Comps											
Dell Technologies	DELL US	349,010	USD	549	1.8x	1.5x	21.2x	18.0x	20.9%	11.3%	n.a.
GE Vernova	GEV US	253,091	USD	950	5.5x	4.8x	39.5x	38.7x	14.9%	15.2%	21.2x
Siemens AG	SIE GY	244,839	EUR	274	2.6x	2.4x	24.4x	21.3x	6.7%	16.0%	3.1x
Schneider Electric	SU FP	195,148	EUR	295	3.8x	3.5x	28.9x	24.3x	9.1%	19.2%	7.0x
Hewlett Packard	HPE US	81,015	USD	61	1.7x	1.5x	16.0x	13.3x	8.1%	15.0%	3.1x
Schindler Holding	SCHN SW	33,719	CHF	257	2.5x	2.4x	25.1x	23.2x	4.3%	13.7%	6.0x
HP Inc	HPQ US	28,893	USD	32	0.5x	0.5x	9.9x	10.1x	0.4%	6.4%	n.a.
Median		195,148		274	2.5x	2.4x	24.4x	21.3x	8.1%	15.0%	6.0x
Global Electronics Comps											
Delta Electronic	2308 TT	156,454	TWD	1,910	6.4x	4.8x	47.3x	30.1x	48.6%	21.7%	16.9x
Hon Hai Precision Industry	2317 TT	113,854	TWD	257	0.3x	0.3x	13.8x	10.9x	31.9%	3.4%	1.9x
Victory Giant Technology	2476 HK	39,074	HKD	224	n.a.	n.a.	n.a.	n.a.	40.4%	#N/A N/A	n.a.
Hubbell Inc	HUBB US	24,241	USD	459	3.5x	3.2x	22.4x	20.1x	5.8%	23.9%	6.2x
Applied Optoelectronics	AAOI US	9,049	USD	107	8.6x	3.2x	106.0x	23.3x	55.8%	18.1%	5.4x
Median		39,074		257	5.0x	3.2x	34.8x	21.7x	40.4%	19.9%	5.8x

Stock Prices as of 2026.09.23; Years stated refers to Calendar Year; General Electric, RTX, and Lockheed Martin are covered by Douglas S. Harned, Ph.D. Subaru is covered by Masahiro Akita. Dell Technologies, Hewlett Packard, and HP Inc are covered by Mark C. Newman. GE Vernova is covered by Sunaina Ocalan. Siemens AG and Schneider Electric are covered by Alasdair Leslie. Delta Electronics is covered by Alex Wang; Bernstein estimates for the seven Japan IT Hardware & Electronics companies, Bloomberg consensus for the rest.

Source: Bloomberg, Bernstein estimates and analysis

DETAILS

Electronic systems rely on active and passive components to regulate power, process signals, and maintain stable operation. Changes in the power source, switching activity, electrical noise, and sudden movements in load current can create ripple, voltage transients, and other disturbances. Active devices, including voltage regulators and switching transistors, detect these changes and adjust power delivery, but their control loops cannot respond instantaneously. Passive components bridge this response gap by storing, releasing, dissipating, transferring, or filtering electrical energy. During a sudden increase in load current, for example, output capacitors initially supply the current shortfall until the regulator adjusts its output. (*Exhibit 2 - Exhibit 4*)

Higher semiconductor performance raises the technical requirements for passive components, but does not necessarily increase component count proportionately. Faster switching, higher operating frequencies, greater integration, and rising power density increase the need for low impedance, low electrical losses, effective noise suppression, compact packaging, and optimized component placement. At the same time, advances in semiconductor integration, passive materials, packaging, and circuit design can consolidate functions and reduce the number of discrete components. The investment opportunity therefore depends not only on unit growth, but also on higher specifications, product mix, component value, and placement within the power-delivery architecture.

- **Passive components:** Passive components cannot provide power gain or independently amplify signals. Instead, their electrical properties allow them to store, release, dissipate, transfer, or filter energy. Capacitors, inductors, and resistors are the three fundamental passive elements, while transformers, ferrites, filters, resonators, and protection devices perform additional functions. Together, these components stabilize power rails, establish voltage and current conditions, suppress noise, protect circuits, and support reliable signal transmission.
- **Capacitors:** Capacitors store energy in an electric field and release or absorb charge as circuit conditions change. They stabilize supply voltage, support transient current, smooth ripple, filter noise, couple signals, and provide timing or energy buffering. Common technologies include ceramic, aluminum electrolytic, aluminum polymer, tantalum, film, silicon, and supercapacitors. Selection depends on capacitance, voltage, frequency response, package size, effective capacitance, ripple current, reliability, operating life, and cost.
- **Inductors:** Inductors store energy in a magnetic field and resist rapid changes in current. They smooth current in power converters, store and transfer energy, suppress high-frequency noise, and form filters and resonant circuits. Wire-wound products typically use copper windings around air, ferrite, powdered-metal, or other magnetic cores, while multilayer, molded, and thin-film structures address more compact applications. Key specifications include inductance, DC resistance, rated and saturation current, core loss, self-resonant frequency, and temperature rise.
- **Resistors:** Resistors limit current and dissipate electrical energy primarily as heat. They set or divide voltage, establish bias conditions, sense current, terminate signal lines, discharge capacitors, and damp unwanted resonance. Common constructions include thick-film, thin-film, metal-film, metal-foil, wire-wound, and metal-oxide resistors. Performance depends on resistance, tolerance, temperature coefficient, power and voltage ratings, pulse capability, long-term stability, and package size.
- **Filters:** Filters pass selected frequency ranges while attenuating unwanted noise or signals. A filter is more accurately described as a circuit function or packaged component than as a fourth fundamental passive element because passive filters are generally constructed from capacitors, inductors, resistors, transformers, ferrites, or resonant materials. Common types include low-pass, high-pass, band-pass, band-stop, electromagnetic-interference, common-mode, ceramic, and acoustic-wave filters.
- **Other passive components:** Transformers transfer AC energy between windings and provide voltage conversion, impedance matching, signal coupling, or electrical isolation. Ferrite beads and common-mode chokes suppress high-frequency noise. Thermistors support temperature sensing and inrush-current limiting, while varistors, fuses, and resettable fuses provide overvoltage or overcurrent protection. Crystals and resonators provide stable frequency references for timing and communications.
- **Active components on the other hand use externally supplied power to control, switch, rectify, process, or amplify electrical signals.** Representative devices include transistors, integrated circuits, voltage regulators, operational amplifiers, processors, memory, and power semiconductors. Diodes are generally classified as semiconductor devices because of

their nonlinear and rectifying behavior, although they do not provide power gain. Active and passive components perform complementary functions and must operate together for an electronic system to function reliably.

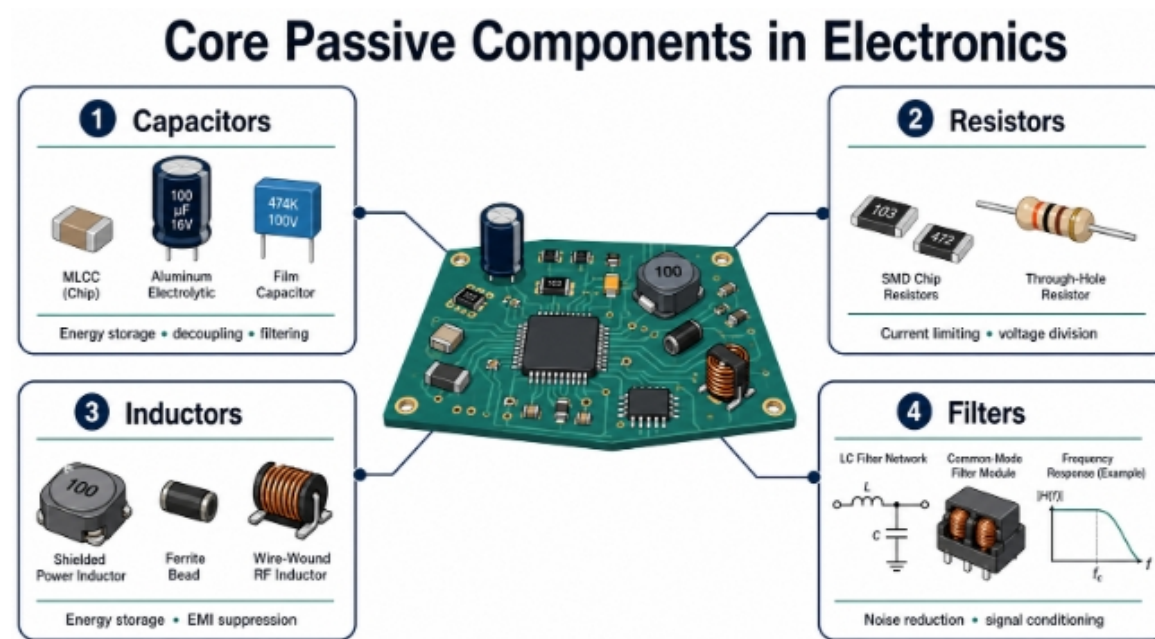
Capacitors are core passive components that stabilize voltage, supply transient current, smooth power-supply ripple, and filter electrical noise. A capacitor stores energy in the electric field between two conductive electrodes separated by an insulating dielectric, with capacitance determined primarily by dielectric permittivity, effective electrode area, and dielectric thickness. Actual performance depends not only on nominal capacitance, but also on equivalent series resistance, or ESR, which determines electrical losses and heat generation, and equivalent series inductance, or ESL, which limits high-frequency response and causes the device to behave increasingly like an inductor above its self-resonant frequency. Voltage rating, effective capacitance under DC bias, temperature stability, ripple-current capability, leakage current, reliability, and operating life are also critical. (*Exhibit 5 - Exhibit 9*)

No single capacitor technology optimizes frequency response, voltage, capacitance, size, reliability, and cost simultaneously, so modern power-delivery networks use a tiered architecture. Silicon capacitors and small MLCCs address the highest-frequency requirements close to the semiconductor package; larger MLCCs and polymer capacitors support board- and converter-level transients; aluminum electrolytic and hybrid capacitors provide bulk capacitance over longer response intervals; and supercapacitors provide short-duration system-level energy buffering. The following analysis focuses on the capacitor technologies most relevant to advanced computing, AI infrastructure, automotive electronics, and high-density power delivery. **The investment implication is that rising processor power density increases the performance requirements and value opportunity across the capacitor stack, although the beneficiaries differ by frequency, placement, system architecture, and technical barrier to entry.**

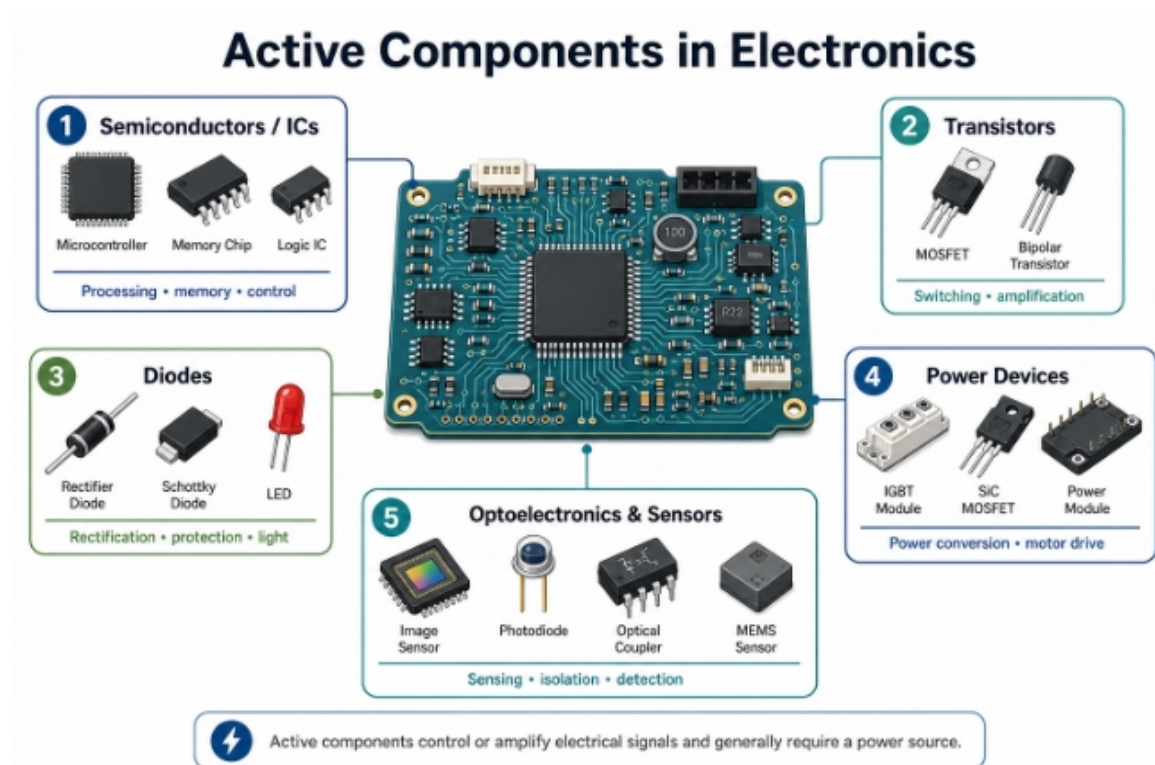
- MLCC (MultiLayer Ceramic Capacitor):** MLCCs combine compact size, low parasitic losses, long operating life, and high-volume manufacturability, making them the primary solution for high-frequency decoupling, noise filtering, signal coupling, and local voltage stabilization near processors, memory, power semiconductors, and other high-speed devices. Their stacked ceramic-dielectric and metal-electrode structure creates substantial effective electrode area and relatively high capacitance within a small package. Performance varies materially by dielectric: Class I products, including COG and NPO, provide greater stability and lower losses but lower capacitance density, while Class II products, including X5R and X7R, offer higher capacitance density but are more sensitive to DC bias, temperature, and aging, meaning nominal capacitance can materially overstate usable capacitance under actual operating conditions. The technological moat is concentrated in advanced rather than mature general-purpose MLCCs, as high-capacitance, high-voltage, automotive-grade, high-temperature, and ultra-small products require uniform thin dielectric layers, precise electrode printing, large layer counts, tightly controlled firing, and consistently high manufacturing yields. AI servers increase both MLCC content and specification requirements because higher processor current, tighter voltage tolerances, denser boards, and faster load transients require more compact, low-impedance capacitance close to processors and power-delivery networks. MLCCs serve a broad range of end markets, but the strongest economics remain concentrated in technically demanding and reliability-sensitive applications.
- Polymer Aluminum Capacitor:** Polymer aluminum capacitors address board- and converter-level requirements where high capacitance, low ESR, strong ripple-current capability, and competitive cost per unit of usable capacitance are more important than minimum ESL. They typically combine an aluminum anode, an aluminum-oxide dielectric, and a solid conductive-polymer cathode, with the polymer replacing the liquid electrolyte used in conventional aluminum electrolytic capacitors to reduce ESR and improve high-frequency and ripple-current performance. They are commonly used for bulk decoupling at the input and output stages of DC/DC converters, voltage-regulator modules, servers, communications equipment, and other power supplies rather than for the highest-frequency decoupling immediately adjacent to the semiconductor. Compared with MLCCs, polymer aluminum capacitors are generally larger and have higher ESL, but provide greater capacitance, limited DC-bias-related capacitance loss, strong ripple-current handling, and attractive usable-capacitance economics. Multilayer architectures offer very low ESR and a low profile, while wound products generally extend into higher capacitance and voltage ranges. Component substitution must also account for regulator-control-loop stability because materially lower ESR can reduce damping and cause ringing or oscillation unless the power supply is designed or compensated for the new impedance profile.
- Polymer Tantalum Capacitor:** Polymer tantalum capacitors combine high volumetric efficiency, stable effective capacitance, and low ESR, making them suitable for compact, performance-sensitive applications requiring predictable bulk capacitance within limited board space. A porous sintered-tantalum anode creates a large internal surface area, while the tantalum-pentoxide dielectric and conductive-polymer cathode support high capacitance in a compact surface-mount package. Unlike many high-permittivity MLCCs, polymer tantalum capacitors experience limited capacitance loss

under applied DC voltage, while the polymer cathode provides lower ESR, better ripple-current capability, and stronger high-frequency performance than conventional manganese-dioxide tantalum capacitors. These characteristics support applications across smartphones, laptops, solid-state drives, networking equipment, industrial systems, AI servers, and DC/DC converters. The principal trade-offs are higher cost, exposure to tantalum pricing and availability, polarity requirements, and a narrower mainstream voltage range than aluminum-based technologies. Polymer tantalum capacitors are therefore most competitive where compact size, stable usable capacitance, low ESR, and reliability justify the premium, but less attractive where lower-cost ceramic or aluminum alternatives can meet the required electrical and packaging specifications.

- Hybrid Capacitor:** Conductive-polymer hybrid aluminum capacitors combine polymer and liquid electrolytes to balance low ESR, strong ripple-current capability, operating-voltage range, temperature performance, and long-term reliability. The polymer improves ESR and high-current performance, while the liquid electrolyte supports higher operating voltages, helps maintain the aluminum-oxide dielectric, and provides the self-healing characteristics associated with conventional aluminum electrolytic capacitors. This combination makes hybrid capacitors particularly suitable for automotive electronics, telecommunications equipment, industrial power supplies, and DC/DC converters requiring relatively high capacitance and resilience under demanding electrical and environmental conditions. Automotive applications are especially important because powertrain, infotainment, advanced driver-assistance, and electronic-control systems must withstand temperature cycling, vibration, high ripple currents, and long operating lives. Hybrid capacitors are generally larger and have higher ESL than MLCCs and silicon capacitors, limiting their suitability for the highest-frequency decoupling, but offer advantages in bulk capacitance, voltage coverage, ripple-current handling, temperature tolerance, and reliability. They therefore occupy the bulk-capacitance and power-conversion layers of the tiered network alongside smaller, higher-frequency ceramic capacitors.
- Silicon Capacitor:** Silicon capacitors address applications where very low parasitic inductance, electrical stability, thin form factor, and proximity to the semiconductor are critical. Semiconductor fabrication processes and three-dimensional structures, including deep trenches, create substantial effective electrode area within a compact footprint, enabling precise electrical characteristics, flexible terminal layouts, very low ESR and ESL, stable capacitance across voltage and temperature, limited aging, and integration within semiconductor packages or package substrates. Placement close to processors, memory, or power semiconductors shortens the current path, reduces parasitic inductance, and improves power integrity as supply voltages decline, processor currents rise, and tolerance for voltage droop narrows. Silicon capacitors can therefore fill the high-frequency gap between on-die capacitance and board-level MLCCs, supporting applications in advanced packaging, radio-frequency systems, optical communications, mobile processors, AI accelerators, and high-performance computing. They are not a broad replacement for MLCCs because higher cost and architecture-specific voltage, capacitance, and packaging constraints make them less economical for large amounts of board-level bulk capacitance. We therefore view silicon capacitors as a high-value complement in advanced systems where power-integrity and packaging benefits justify the premium, potentially increasing capacitor value per package even if selected designs require fewer discrete board-level components.
- Super Capacitor:** Supercapacitors, also known as ultracapacitors or electrochemical double-layer capacitors (EDLC), provide high power density, rapid response, and long cycle life for short-duration energy buffering. High-surface-area porous-carbon electrodes and an electrolyte store charge primarily at the electrode-electrolyte interface, enabling substantially greater capacitance and energy storage than conventional capacitors. Applications include pulse-power delivery, regenerative-energy capture, ride-through support, rapid power buffering, and short-duration backup. Their energy density nevertheless remains well below that of lithium-ion batteries, limiting their suitability for sustained backup and positioning them alongside batteries, generators, and UPS systems rather than as direct replacements. AI data centers could create a new demand pool as volatile workloads cause rapid changes in power consumption, with supercapacitors absorbing or supplying energy during short transients, smoothing demand on upstream infrastructure, and reducing shallow cycling and transient stress in battery-based UPS systems. High-voltage DC distribution could provide an additional opportunity, although individual cells operate at relatively low voltages and require series connection, balancing, monitoring, and power-conversion circuitry. We therefore view supercapacitors as an emerging energy-buffering layer for high-density AI infrastructure, with adoption dependent on system cost, footprint, thermal management, reliability, balancing requirements, and integration with existing UPS and power-conversion architectures.

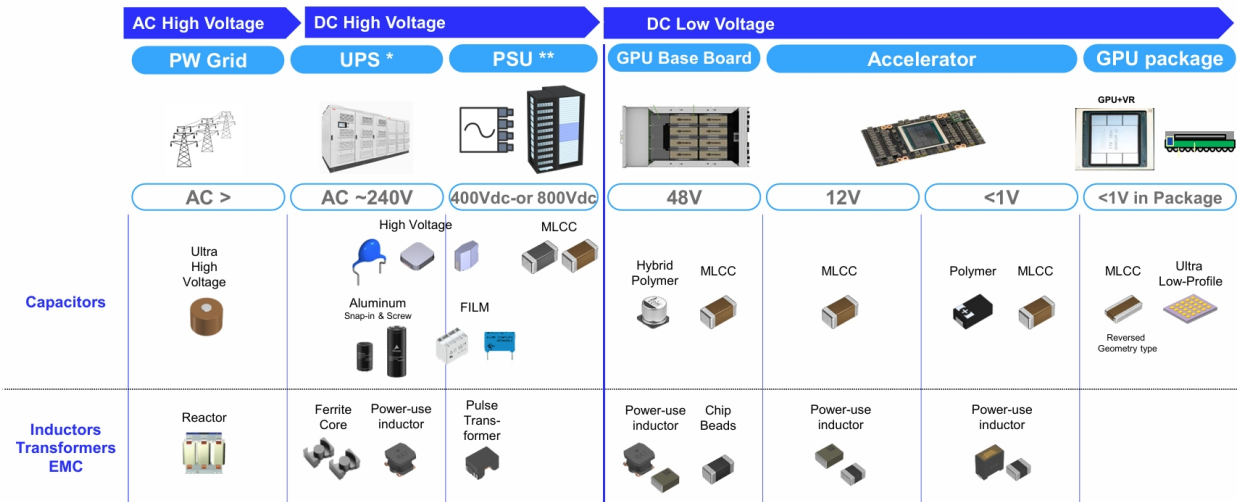
EXHIBIT 2: **Passive components are essential in stabilizing electrical power in a device**

Source: Bernstein analysis

EXHIBIT 3: **Active components consume power to change a signal dynamically**

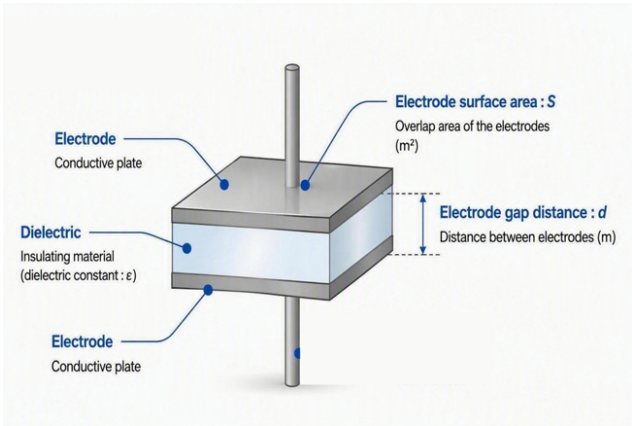
Source: Bernstein analysis

EXHIBIT 4: **Passive components are optimized for different voltage domains and functions across the power rail to support stable system operation**



Source: TDK, Bernstein analysis

EXHIBIT 5: **Basic construction of Capacitor**



Source: Bernstein analysis

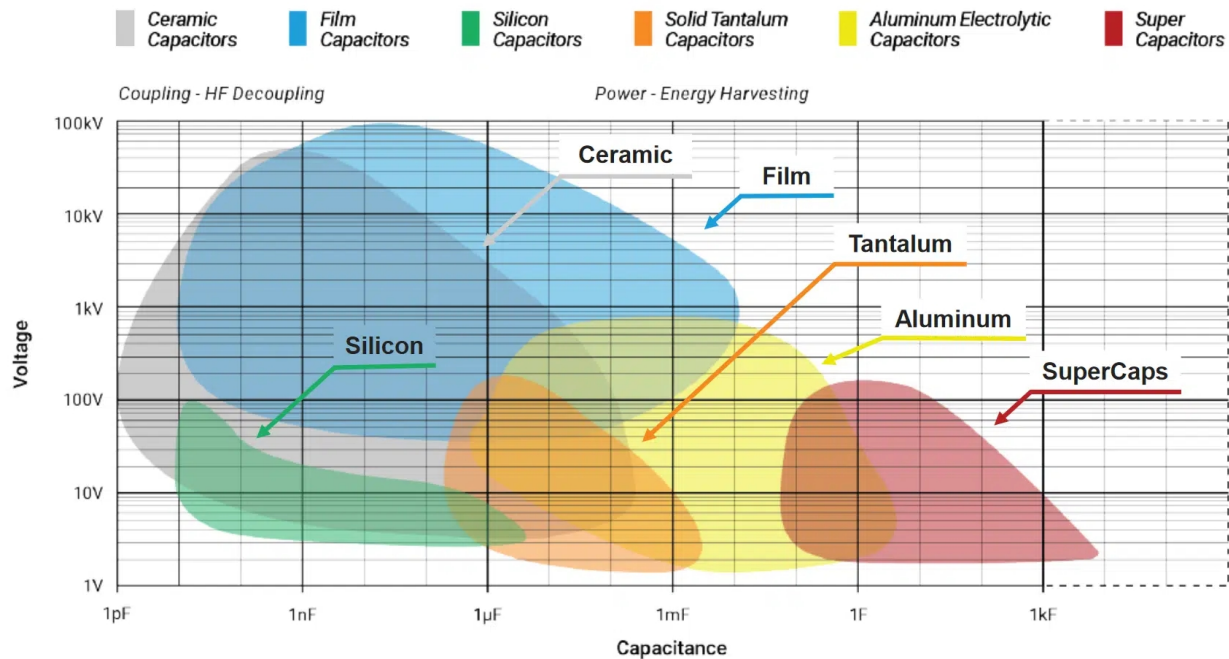
EXHIBIT 6: **Material, Area, and Distance determines Capacitance**

$$C = \frac{\epsilon S}{d}$$

- S: Electrode surface area
- d: Electrode gap distance
- ε: Dielectric constant of dielectric

Source: Bernstein analysis

EXHIBIT 7: Capacitors serve different application depending on Voltage and Capacitance



Source: Passive components blog, Bernstein analysis

EXHIBIT 8: Different capacitors have varying operational requirements and hence best suited for different use cases

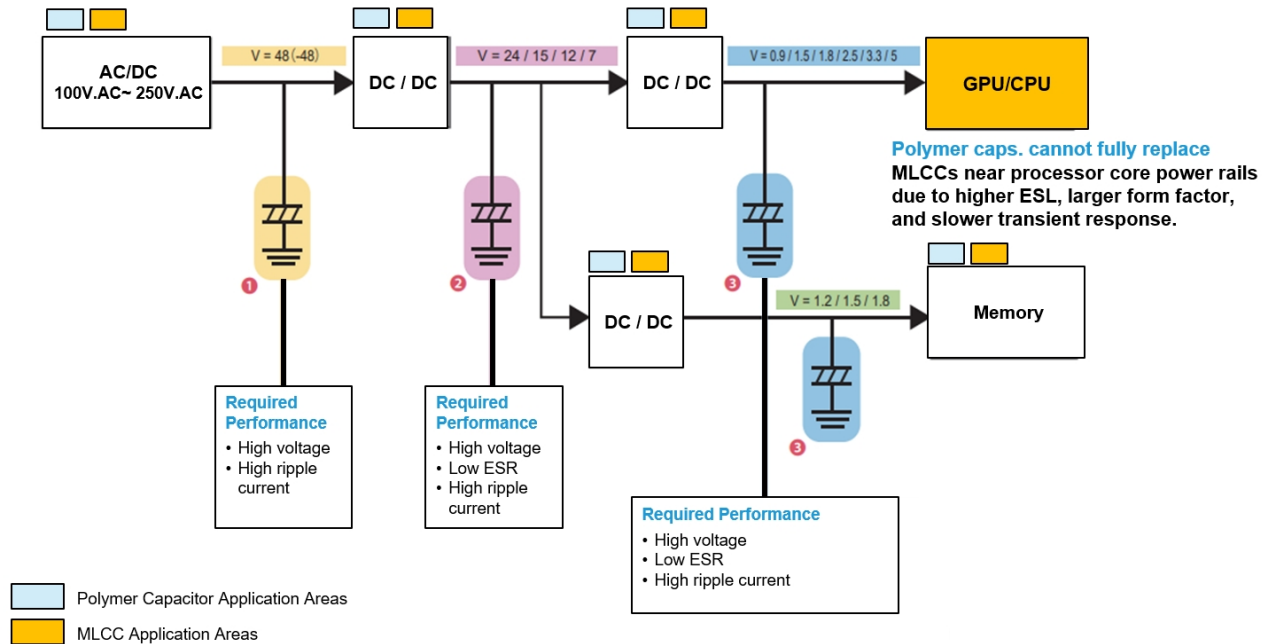
	Multi Layered Ceramic Capacitor (MLCC)	Conductive Polymer Aluminum Capacitor	Conductive Polymer Tantalum Capacitor	Hybrid Capacitor
Capacitance Density	Low-Mid (pF ~ 100uF+)	Medium (2.2uF ~ 560uF)	High (3.9uF ~ 1,500uF)	Very High (3.9uF ~ 2,700uF)
Voltage	Best (2.5V ~ 3kV+)	Medium (2V ~ 35V)	Medium (2V ~ 35V)	High (25V ~ 80V)
DC Bias	High	Negligible		
ESR(*)	Best (1mΩ ~ 30mΩ)	Good (3mΩ ~ 40mΩ)	Medium (5mΩ ~ 300mΩ)	Medium (20mΩ ~ 120mΩ)
ESL(*)	Best (50pH ~ 2nH)	Good (2 ~ 10nH+)	Good (2 ~ 10nH+)	Good (2 ~ 10nH+)
Size	Smallest	Small-Medium	Small-Medium	Medium-Large
High-Frequency	Best	Good	Medium	Medium
Heat resistance	Good (crack-risk)	Good	Good	Best
Tech complexity	High	Relatively limited differentiation among suppliers		

(*) **ESR**: Internal resistance; lower ESR means less heat and better current handling
ESL: Internal inductance; lower ESL means faster response to high-frequency noise

Strong Area

Source: Bernstein analysis

EXHIBIT 9: Capacitors across power rails: MLCCs and capacitors are deployed based on their respective functions



Source: Panasonic, Bernstein analysis

MLCCs are the biggest buzzword in the capacitors, characterized by a high degree of technological differentiation.

MLCCs utilized a multilayer structure where ceramic dielectric layers and internal Nickel electrodes are stacked in hundreds to thousands of layers. This structure enables extremely low ESL and ESR, while delivering high capacitance within a compact footprint. Increasing capacitance without enlarging component size requires thinner dielectric layers, finer electrode formation, and a greater number of stacked layers, creating significant technological barriers to entry. As a result, only a handful of manufacturers possess the process technology and production know-how necessary to commercialize leading-edge MLCC products at scale. (*Exhibit 10, Exhibit 11*)

While multiple capacitor technologies are used across power rails, MLCCs become increasingly critical closer to semiconductor chips. Processors, including CPUs, GPUs, and MCUs, generate rapid current transients during switching operations, making voltage stability a key performance requirement. Due to their compact size and ultra-low ESL/ESR characteristics, MLCCs are highly effective at suppressing high-frequency voltage noise and providing localized charge near the processor. Their importance is even greater in high-performance computing systems, where limited board space and rising power densities require capacitors that combine small form factors with high capacitance and excellent high-frequency performance. **Consequently, MLCCs have become indispensable components in modern electronic systems, and their strategic importance is increasing further with the expansion of AI infrastructure, EVs, and ADAS.** (*Exhibit 12, Exhibit 13*)

The variety of MLCCs are used across industries based on power, temperature and space requirements. General-purpose MLCCs are primarily used in electronic appliances, accounting for most unit volumes. Compact and high integrated devices, such as smartphones require smaller and more advanced MLCCs due to limited board space. Computing and automotive applications use high-end MLCCs that provide high capacitance and reliable performance under demanding operating conditions. Automotive MLCCs, in particular, must pass stringent automotive-grade qualification tests, creating an additional barrier to entry. Most recently, AI data centers have emerged as a key market for cutting-edge MLCCs, as GPUs and AI accelerators require compact, high-capacitance products placed nearby to buffer rapid current fluctuations and maintain stable voltage under severe space and thermal constraints. (*Exhibit 14, Exhibit 15*)

- **Volume based:** Used in high-volume consumer electronics such as home appliances, TVs, and general-purpose devices.

These products typically have relatively standard capacitance, voltage, and reliability requirements. Cost competitiveness, production scale, and stable yields are more important than cutting-edge performance.

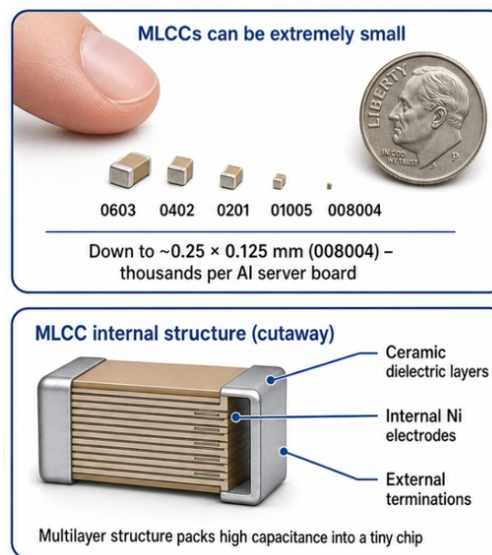
- **Premium/auto grade:** Used in automotive and computing applications. Automotive-grade MLCCs must operate reliably under high-voltage, high-temperature, and other harsh conditions, while meeting stringent qualification requirements. They are primarily used in powertrains, ADAS, and core automotive computing systems. Computing applications require compact MLCCs with strong high-frequency characteristics and fast transient response near chips, including CPUs, storage devices, and power-delivery circuits.
- **Cutting edge:** Designed for high-performance computing chips such as GPUs and AI accelerators, these cutting-edge MLCCs deliver exceptionally high capacitance within a compact footprint. They support power decoupling and stabilization during rapid load fluctuations, while enabling more capacitors to be mounted within limited board space. **Recent advances, including 0201-inch 10 μ F, 0402-inch 47 μ F, and 0603-inch 100 μ F products, are reportedly being adopted in the latest AI server platforms.**

EXHIBIT 10: **Multi-Layered Ceramic Capacitors are one class of capacitors that uses the dielectric properties of ceramic**



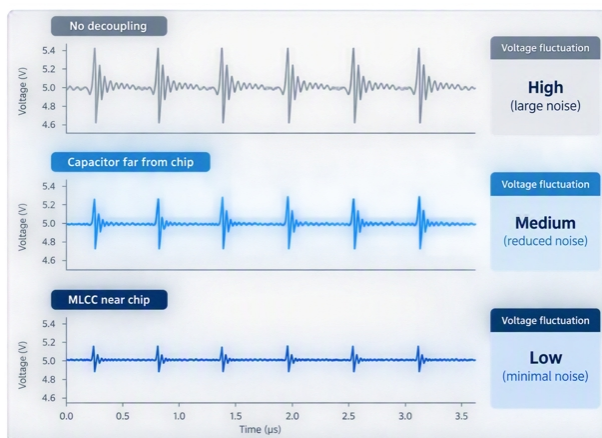
Source: Murata, Bernstein analysis

EXHIBIT 11: **Miniaturization is complex and essential- Technological moat and know how offers differentiation**



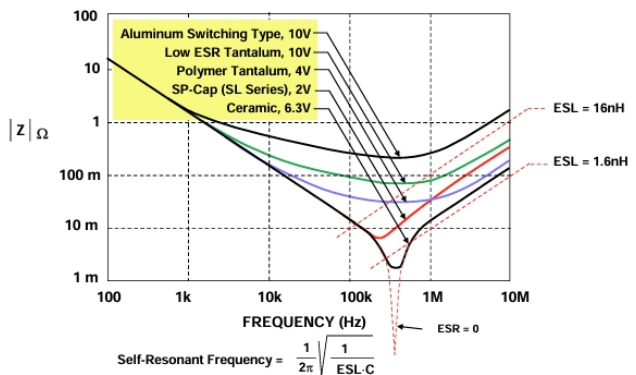
Source: Murata, Bernstein analysis

EXHIBIT 12: **MLCCs ensure stable power delivery near the chip through small size and low ESR/ESL**



Source: Bernstein analysis

EXHIBIT 13: **Low ESL and ESR enable MLCCs to quickly buffer rapid current fluctuations near the chip, helping maintain stable voltage during high-speed operation.**



Source: Analog Devices, Bernstein analysis

EXHIBIT 14: **MLCC applications across a wide range of end markets**

End Product	Number of MLCCs per unit	Type	Function
Smartphone	500	Low-end	Power and display
Edge Device	500	Mid-end	IoT, Compute, Sensor
Smartphone (Flagship)	1,000	Mid-end, High-end	Camera, Compute
Tablet/Laptop	1,500	Mid-end	Compute
ICE Vehicle	1,000	Auto-grade	Infotainment, Power
Robots	5,000 – 10,000	Auto-grade, High-end	Actuator, Sensor, Compute
EV/ADAS Vehicle	10,000 – 30,000	Auto-grade, High-end	ADAS, Sensor, Power, Compute
Military Drones	10,000 – 30,000	High-end	Radar, Sensors, EW, Compute
Conventional Server	5,000	High-end	CPU, Power, Storage
AI Server	30,000 – 50,000	Cutting-edge	GPU, HBM, CPU
Satellite/Spacecraft	Mission-dependent (10,000 – 100,000)	Space-grade	Avionics, Network, Power, Compute

Source: Bernstein analysis

EXHIBIT 15: **Automotive requires high-end MLCCs qualified to AEC-Q200 standards for harsh operating conditions**

Grade	Temperature Range	Type
Grade 0	-50°C to +150°C	Engine Cylinder Head, Transmission, Turbocharger
Grade 1	-40°C to +125°C	Power train, Electric Control Unit, Break system, Engine compartment
Grade 2	-40°C to +105°C	Cabin areas with elevated internal temperature
Grade 3	-40°C to +85°C	General passenger cabin
Grade 4	0°C to +70°C	Non-automotive, limited temperature variation

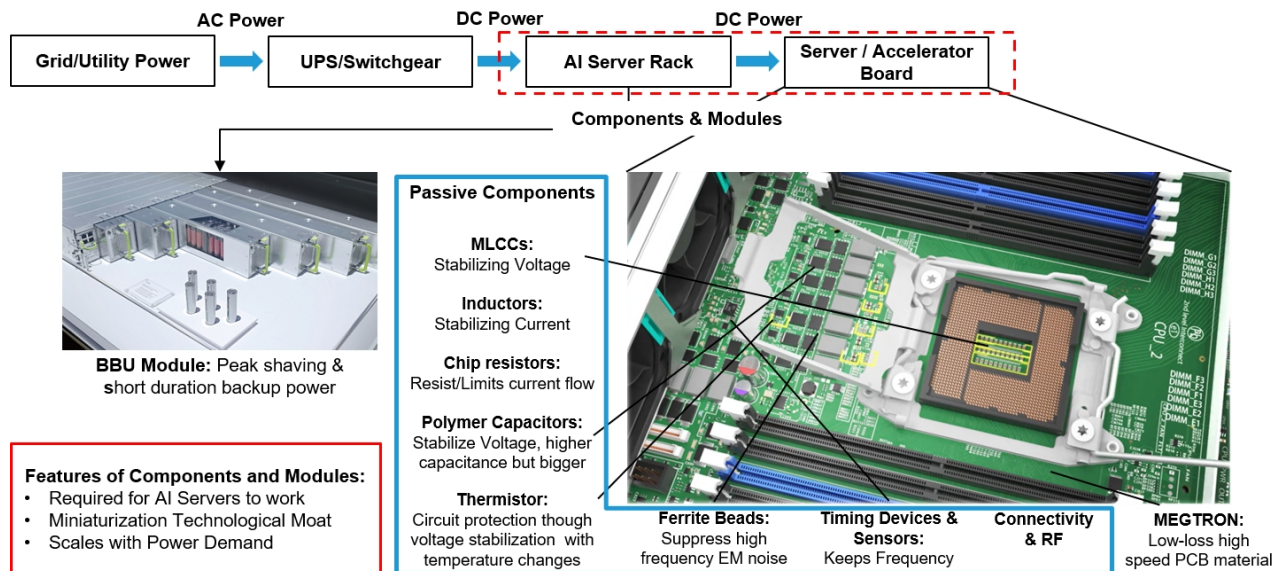
Source: AEC, Bernstein analysis

Strict data center specifications has asymmetrically benefited high-end/cutting-edge MLCCs. MLCCs in AI server PCBs must withstand high power loads, rapid current fluctuations, and high-frequency switching environments within increasingly constrained board space, while ensuring reliable operation at elevated temperatures. Space constraint dictates MLCCs as the choice compared to other capacitors of different materials. As AI Data Center power grows by 19% CAGR till 2030E, we expect cutting-edge MLCCs demand to grow by 33% CAGR to USD27.7Bn by 2032E.

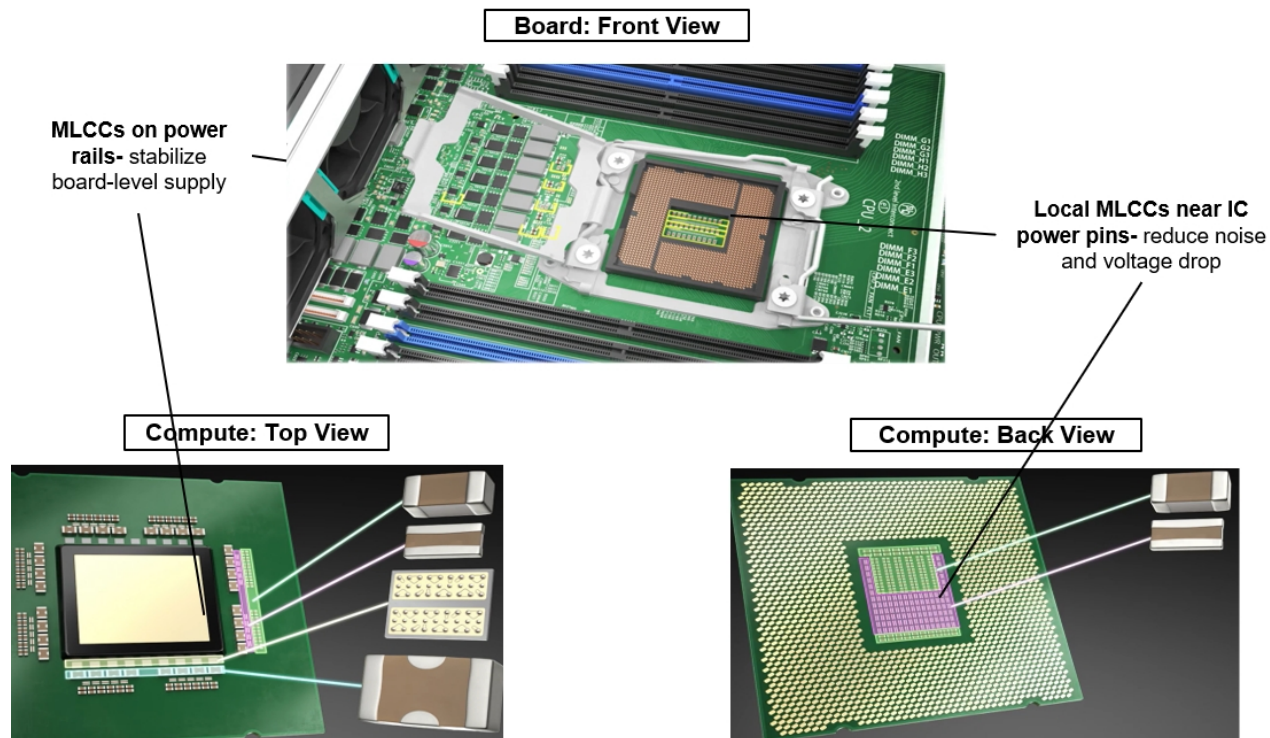
- **AI advancement pushes higher power requirements within tight space constraints within the circuit board, requiring a growing number of MLCCs with diverse capacitance values and package sizes to optimize power delivery.** MLCCs are placed near key active areas such as around and beneath the GPU/AI Chips. MLCCs are uniquely positioned to support high-frequency power delivery, suppress noise across a wide frequency range, and provide flexible design options through a broad range of capacitance values and package sizes, driving higher MLCC content per system. In addition, MLCCs must ensure high reliability, as AI servers and accelerator modules operate 24/7 under elevated thermal conditions where failures are unacceptable. (*Exhibit 16, Exhibit 17*)

- **Space constraint especially underneath and around the GPU dictates MLCC as the preferred use case over other forms of capacitors.** While polymer capacitors and hybrid capacitors have higher capacitance, the materials are not the densest. The ceramic properties enable more efficient use of limited PCB space. More importantly, higher capacitance within a smaller footprint allows a greater number of capacitors to be placed close to processors and other power-sensitive components, improving power integrity and overall system density. Historically, MLCC miniaturization was driven by compact edge devices such as smartphone. However, AI servers represent a game changer, as GPUs and AI accelerators combine exceptionally high power consumption with rapid and substantial fluctuations in current demand, all within extremely constrained board space. (Exhibit 18 - Exhibit 22)
- **The expansion of AI investment is driving new server deployments, driven by advanced AI servers and increasing power demand.** Data center power demand is expected to grow at 19% CAGR over the next 7 years driven by more advanced AI servers that require more power. As chip performance and computing capacity have increased significantly since the V100 generation, power consumption per accelerator has risen sharply, driving a corresponding increase in MLCC demand. (Exhibit 23 - Exhibit 25)
- **Data Center MLCCs will grow by 22% but cutting edge MLCCs for AI DC will grow more at 43% CAGR.** We expect the number of MLCC to increase at 22% CAGR, driven by cutting-edge MLCC for High-end AI data center. In addition, moderate price increases for cutting-edge MLCCs and a higher mix of cutting-edge should drive overall data center MLCC ASP growth. Driven primarily by high-end AI server MLCCs, we estimate the data center MLCC market will expand from roughly US\$2.8Bn in 2024 to around US\$27.7Bn by 2032E, with growth accelerating toward 33% YoY in the outer years. (Exhibit 26 - Exhibit 31)

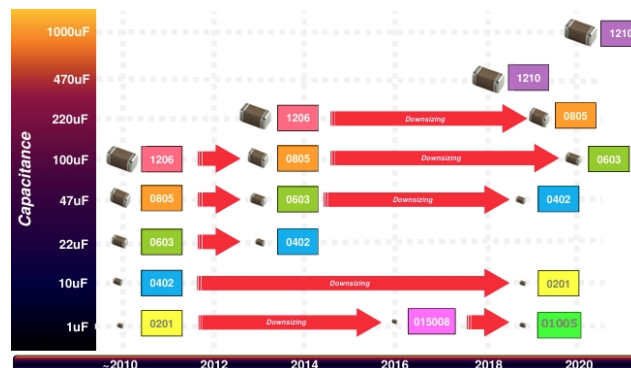
EXHIBIT 16: AI Servers Pull More Value into Power-Critical Components and Modules



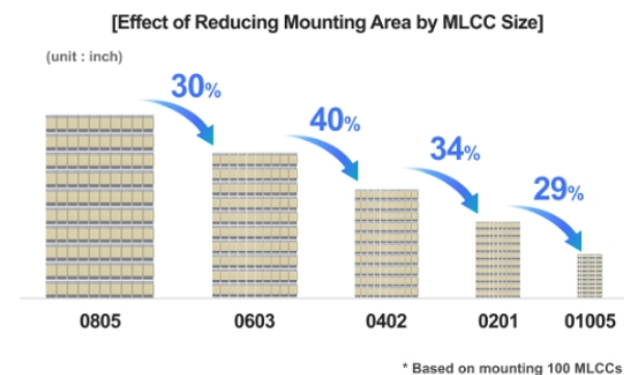
Source: Company websites, Bernstein analysis

EXHIBIT 17: **MLCCs provide miniature power stabilization inside AI server boards**

Source: Murata, Bernstein analysis

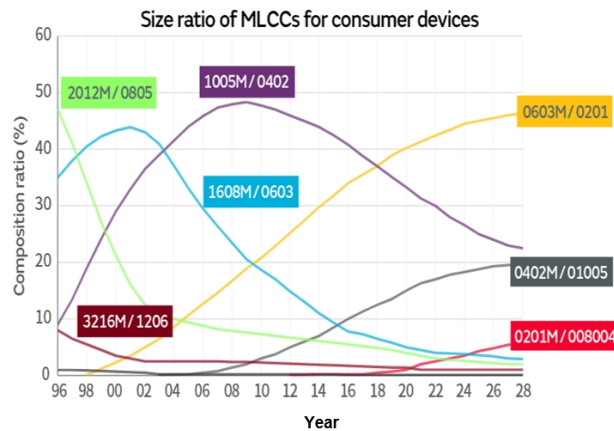
EXHIBIT 18: **Trend is toward smaller MLCCs with the same Capacitance**

Source: Murata, Bernstein analysis

EXHIBIT 19: **Same Capacitance, Smaller MLCCs, More Board Space**

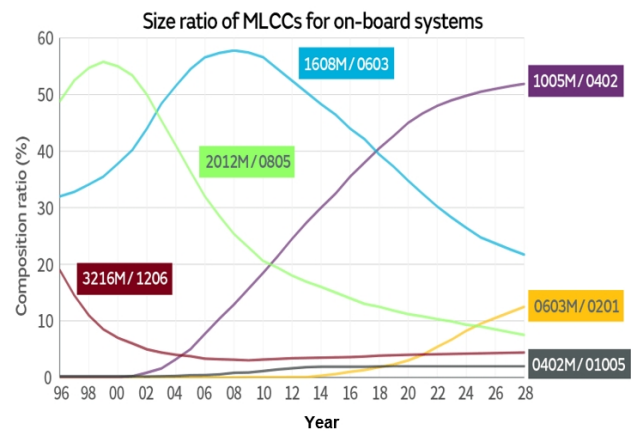
Source: SEMCO, Bernstein analysis

EXHIBIT 20: **Murata's proportion of MLCCs are moving towards those with smaller dimensions**



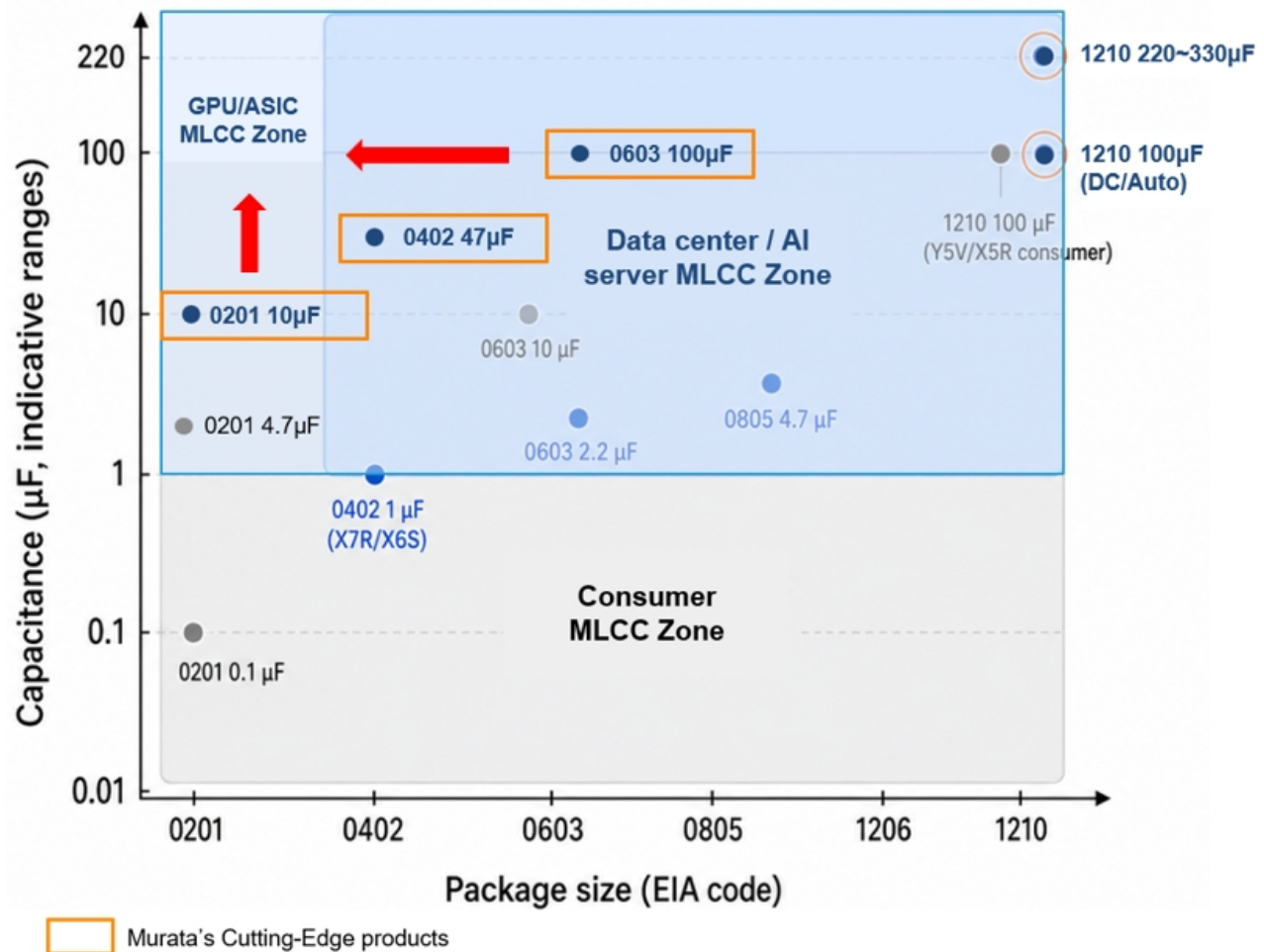
Source: Murata, Bernstein analysis

EXHIBIT 21: **Similar for automotive, there is a constant shift towards smaller capacitors**

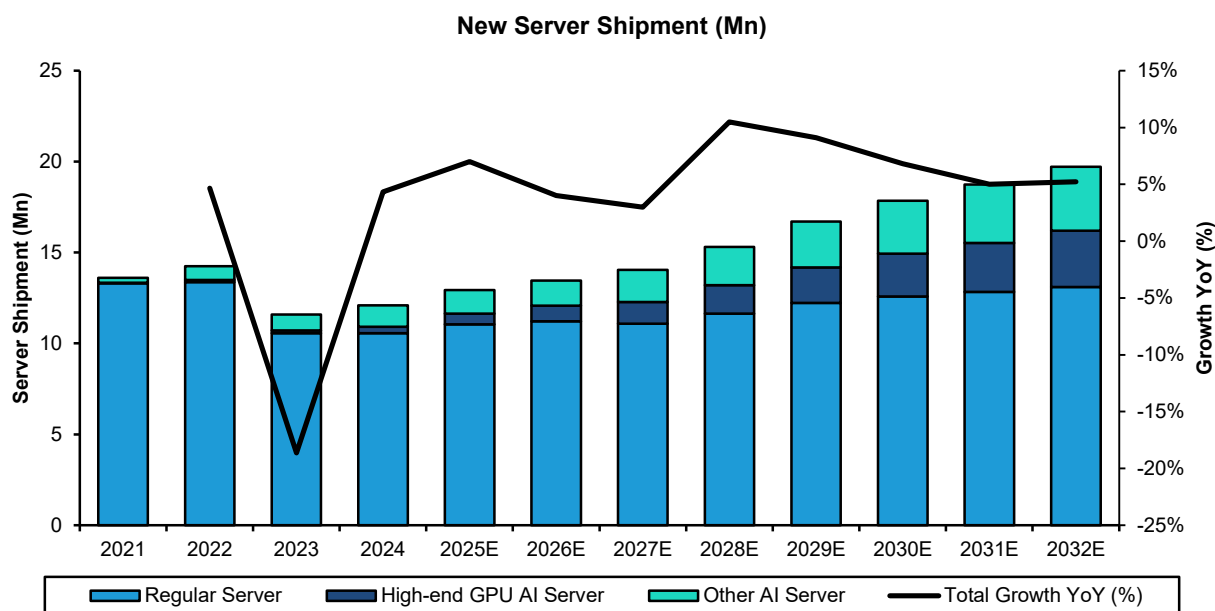


Source: Murata, Bernstein analysis

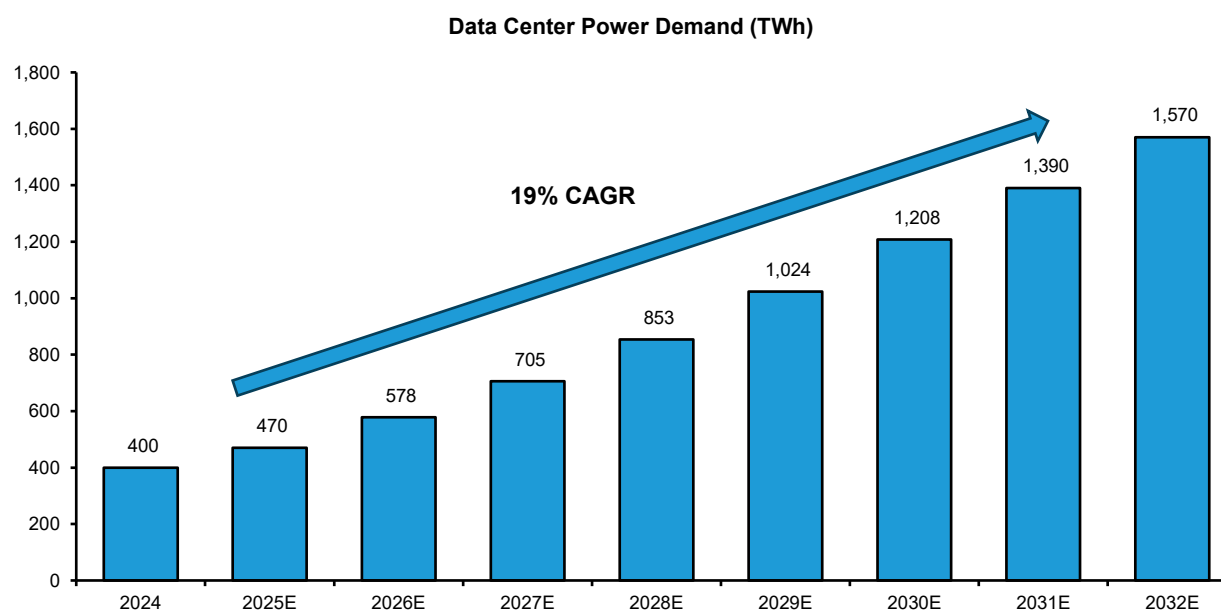
EXHIBIT 22: **AI Servers require higher capacitance in tight spaces within the circuit board**



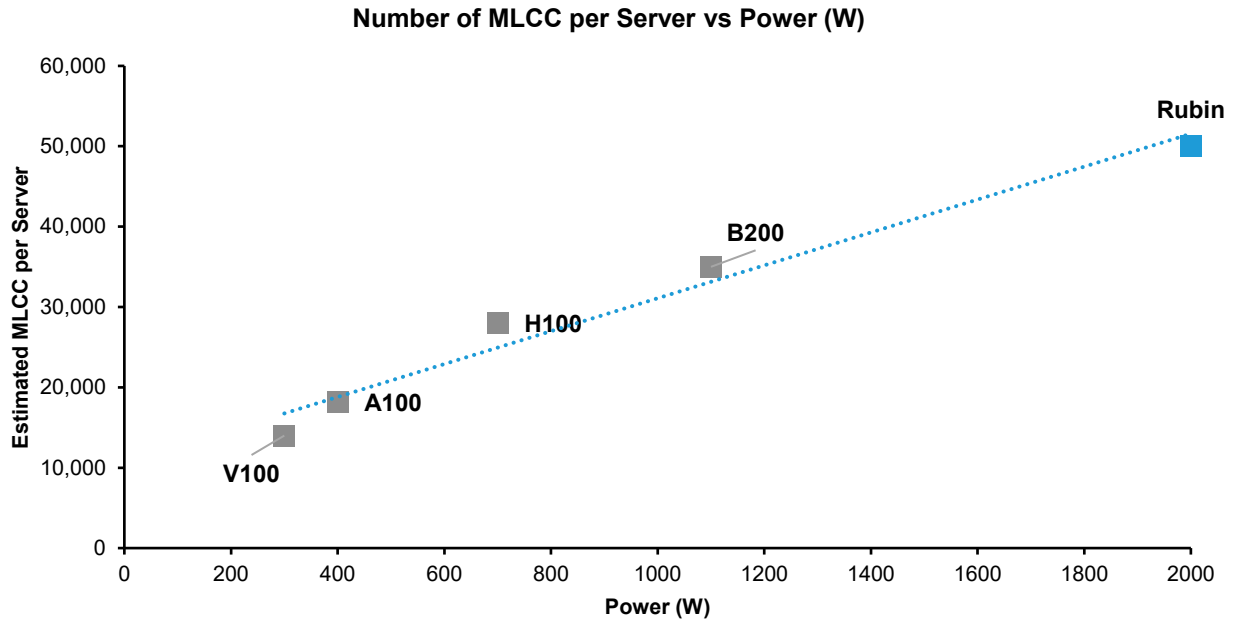
Source: Bernstein analysis

EXHIBIT 23: **AI servers are expected to lead growth in overall data center market**

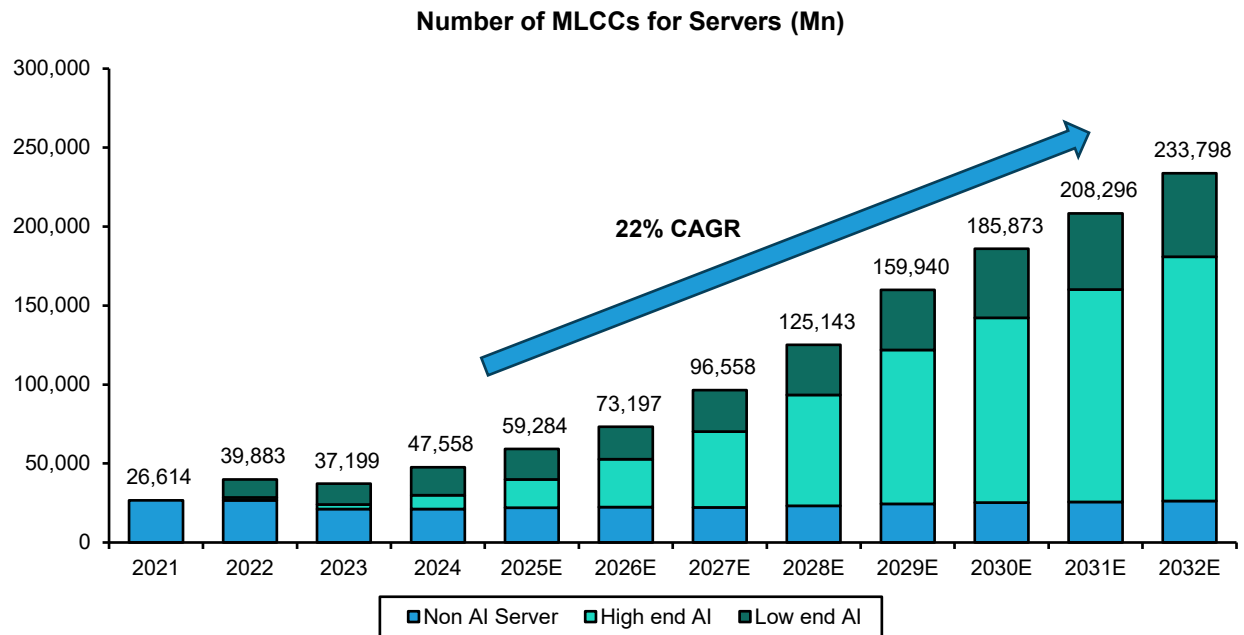
Source: Gartner, Bernstein Asia Tech Hardware Team, Bernstein estimates and analysis

EXHIBIT 24: **Data center power demand is expected to grow at ~19% CAGR, driven by AI servers that consume more power**

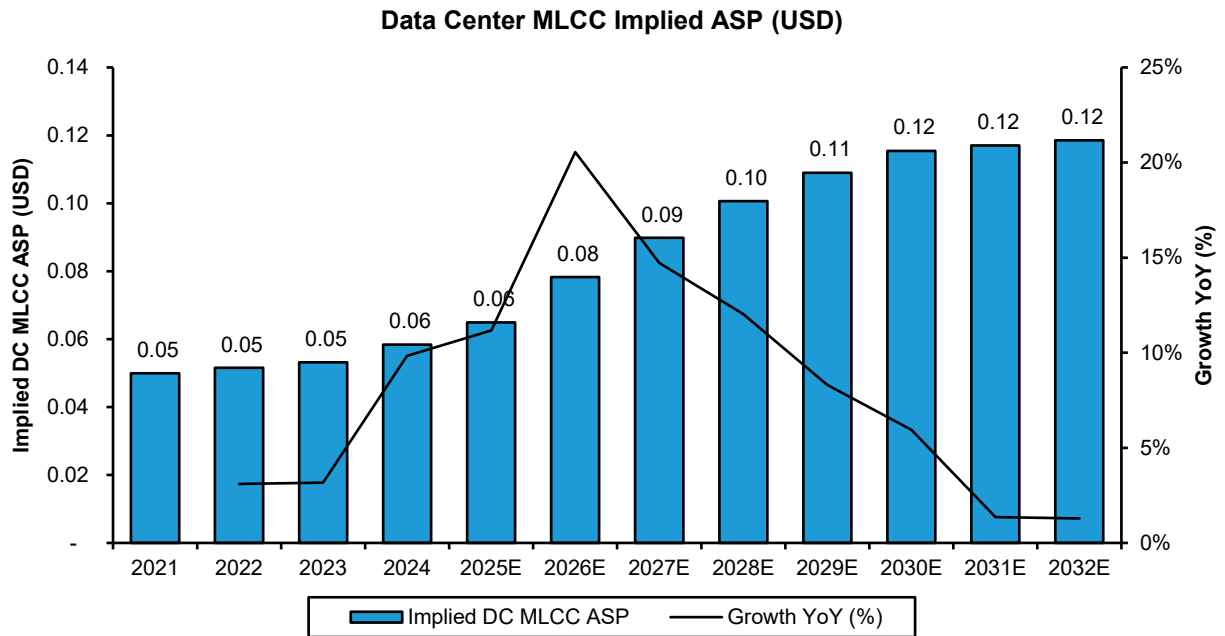
Source: IAE, Bernstein estimates and analysis

EXHIBIT 25: **High-Power AI Servers Need More Premium MLCC Content**

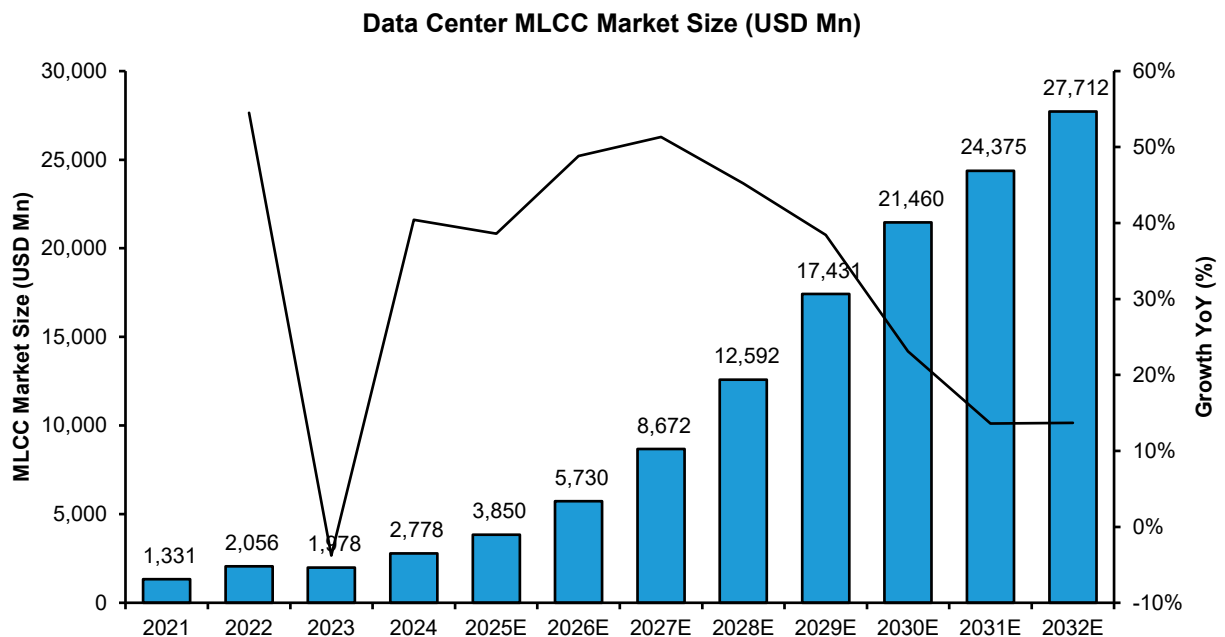
Source: Bernstein estimates and analysis

EXHIBIT 26: **Rising server power and AI drive nonlinear growth in MLCC demand**

Source: Company reports, Bernstein estimates and analysis

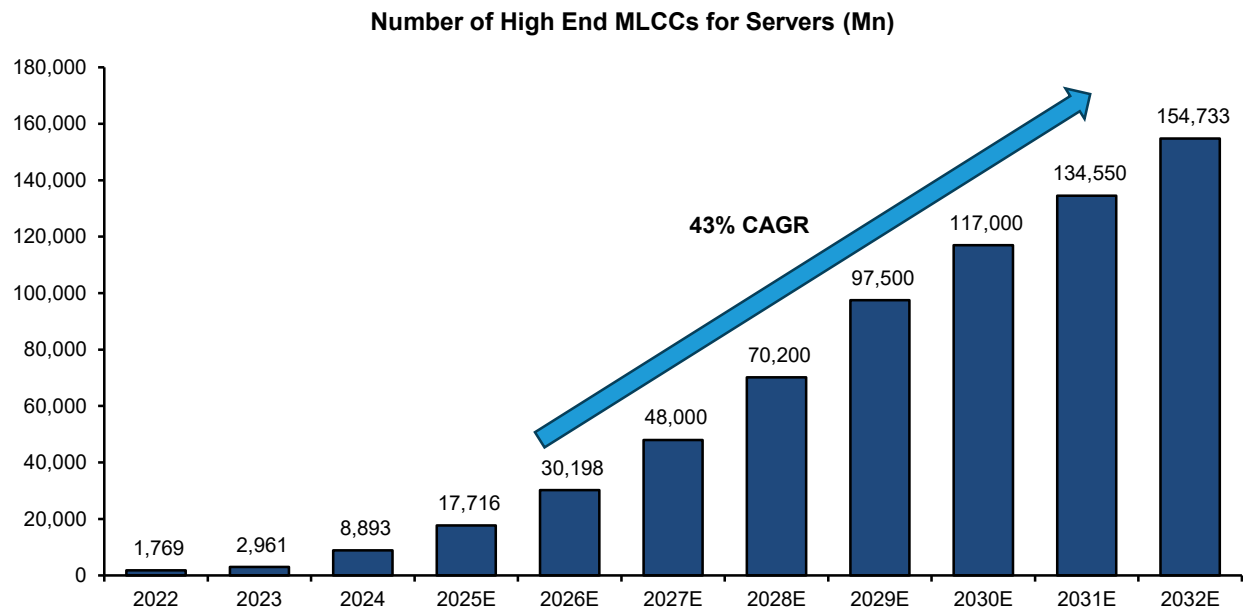
EXHIBIT 27: **MLCC ASP of Data center will rise gradually, as High-End AI MLCC mix increases**

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 28: **Data center MLCC growth accelerates as each incremental AI server requires more power hardware**

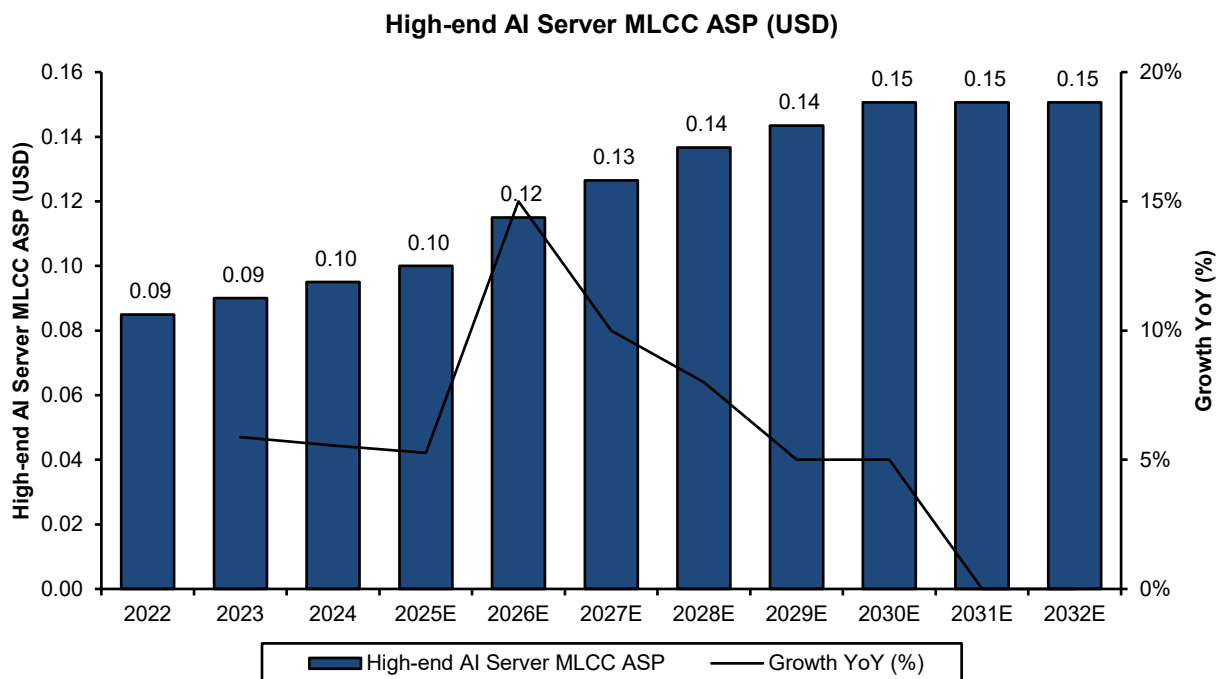
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 29: **Cutting Edge MLCC Demand should outgrow general electronics demand, growing at +43% CAGR in volume**

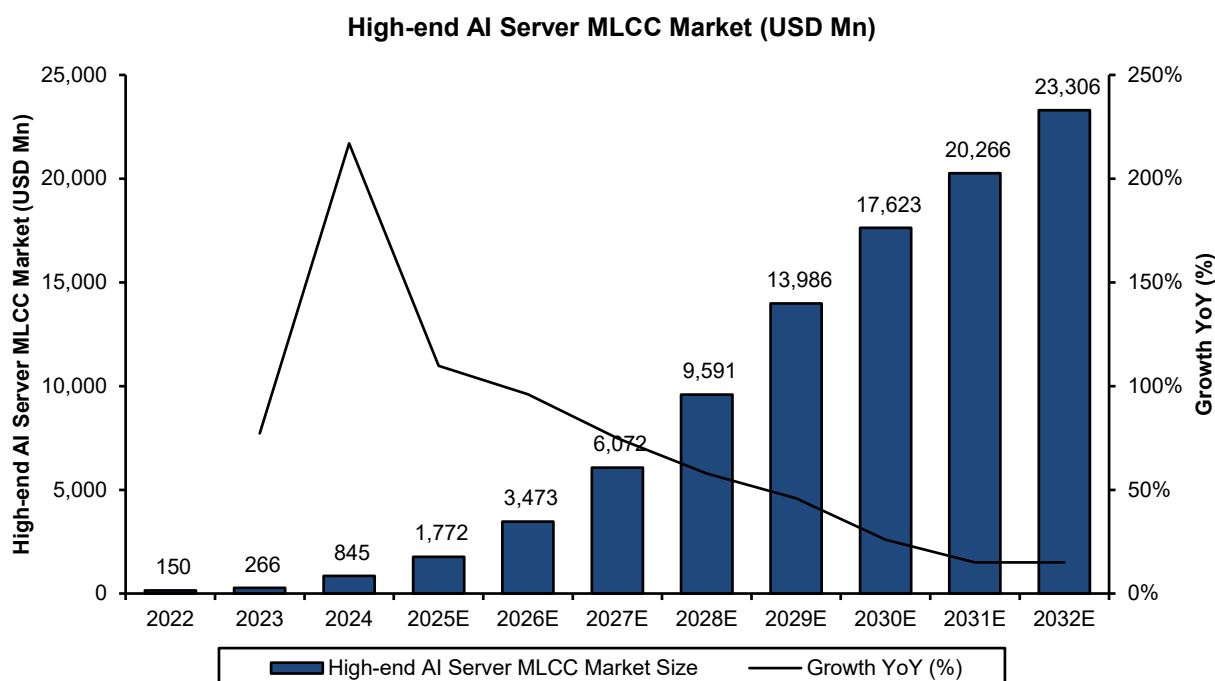


Source: Company reports, Bernstein estimates and analysis

EXHIBIT 30: **Tight supply-demand dynamics in high-end MLCCs are expected to drive gradual ASP expansion**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 31: **High-end AI Server MLCC market is expected to grow at 51% from US\$845Mn to US\$23.3Bn by 2032E**

Source: Company reports, Bernstein estimates and analysis

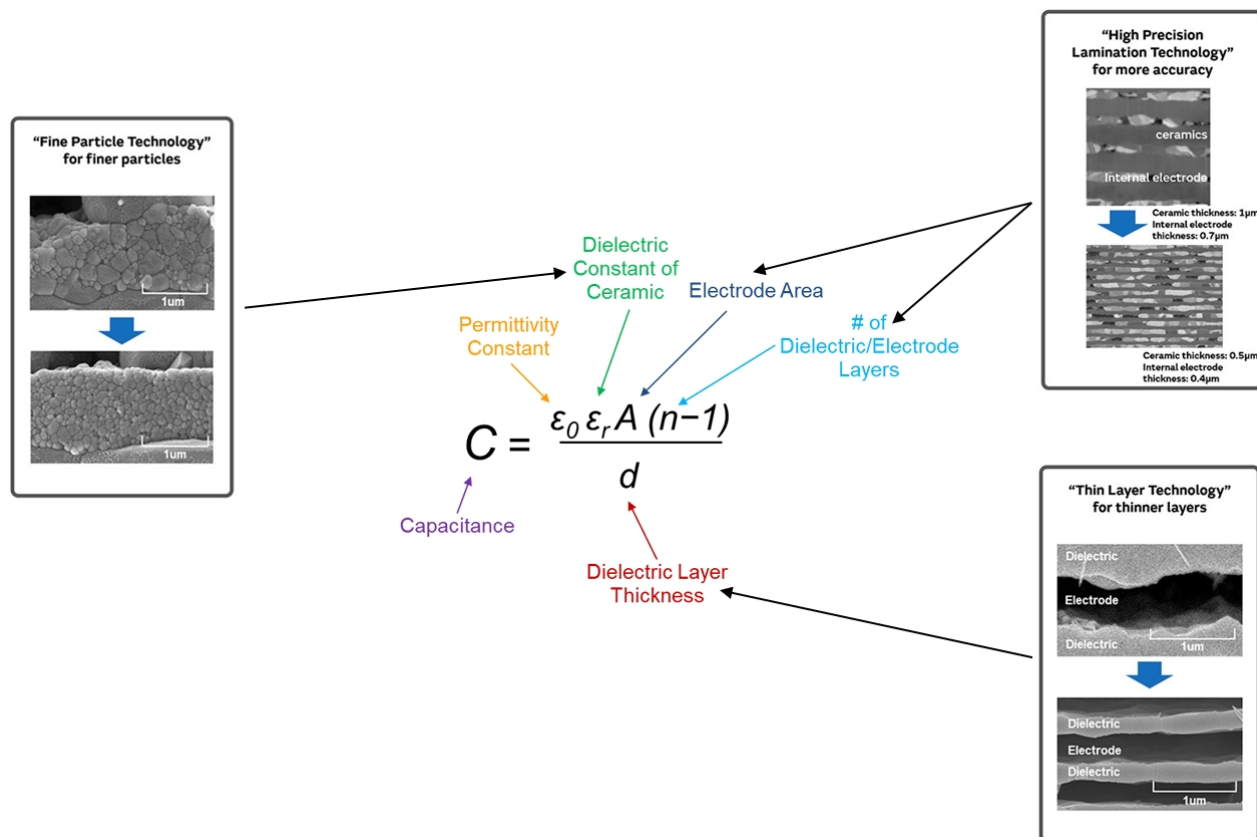
Difficulty in producing cutting-edge MLCCs would see supply tightness at least till 2030E. Producing cutting-edge MLCC for AI Data Center requires miniaturization, material and manufacturing specialty. These specialties are complex and only few companies, Murata and Samsung Electro-Mechanics (SEMCO) - not covered - could produce them effectively at high yields. Capacity is estimated to grow at 50% CAGR attempting to catch up to demand growth at 30% CAGR through 2031E. As such, supply tightness will not be resolved at least in the next 5 years, resulting in potential further price uplifts.

- The key challenge in MLCC manufacturing is achieving higher capacitance within a given case size and rated voltage.** Based on the capacitance formula, increasing capacitance involves thinning the dielectric layers, increasing the layer count, and using materials with higher dielectric constants. However, the real technological barrier lies in improving capacitance while maintaining heat resistance, stability, and long-term reliability under demanding operating conditions, all while maintaining high production yields. (*Exhibit 32*)
- The MLCC manufacturing process is highly complex.** It spans from material formulation and mixing, the precise printing of ultra-thin dielectric and electrode layers, which are then stacked into several hundred layers, dicing the stacked sheets into individual chips, and co-firing them under carefully controlled temperatures and conditions. Manufacturing becomes increasingly difficult for high-end products, as they require thinner layers, superior material uniformity, and a higher layer count, creating a high-end market dominated by a select few established players. **Murata is the undisputed market leader in MLCCs, supported by its strong technological capabilities, industry-leading production capacity, and its long history of proven reliability.** (*Exhibit 33*)
- Premium MLCCs used in computing and automotive applications are expected to see rapid growth, driven by rising AI-related demand and increasing electronic content in vehicles.** Leading premium suppliers, including Murata, SEMCO (Not covered), and Taiyo Yuden (Not covered), are expanding capacity to meet rising market demand. We expect the premium MLCC market, encompassing computing and mobility applications, to grow at a 16% CAGR, with the data-center segment within this market growing faster at a 27% CAGR. (*Exhibit 34, Exhibit 35*)
- AI DC MLCC Capacity is expected to grow at 50% CAGR attempting to catch up with demand growth at 42% CAGR through 2031E.** These cutting edge MLCCs are those high capacitance of 47-100 microfarad with the standard size of 0402-0603. We believe only Murata and SEMCO (Not covered) can produce them at scale and commercialize effectively. Taiyo Yuden (Not covered) lags behind but is still able to produce these cutting edge capacitors. As the AIDC demand for

MLCC increased rapidly, these companies are attempting to catch up despite already having a supply demand gap today. We expect Murata's cutting edge MLCC volume to grow at 60% CAGR through 2031E on faster expansion and production know how while SEMCO (Not covered) and Taiyo Yuden (Not covered) cutting edge MLCC likely to grow at 35% and 25% respectively. Total AIDC MLCC is hence expected to grow at 50% CAGR. (Exhibit 36)

- **Supply tightness is here to stay with potential price increases.** Even as supply grows at 50%, we expect cutting edge MLCC demand will continue growing at 42% CAGR on higher end chips and more power required on the same restricted space in a circuit board. Therefore, we do not see supply catching up with demand anytime soon. As a result, we can expect continued price uplift in the primary market of around 15-30% in the near term and moderate to 5%-10% in the longer run. (Exhibit 37 - Exhibit 39)

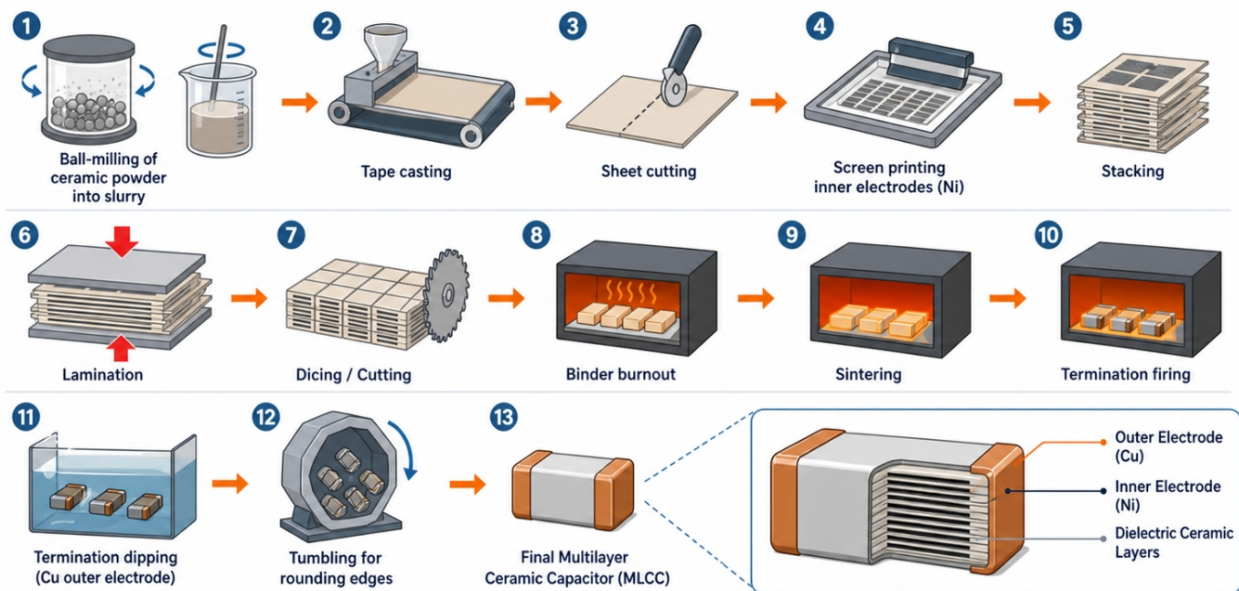
EXHIBIT 32: Murata's high performing MLCCs are based on technological advantages in the materials and manufacturing processes



Source: Company reports, Bernstein analysis

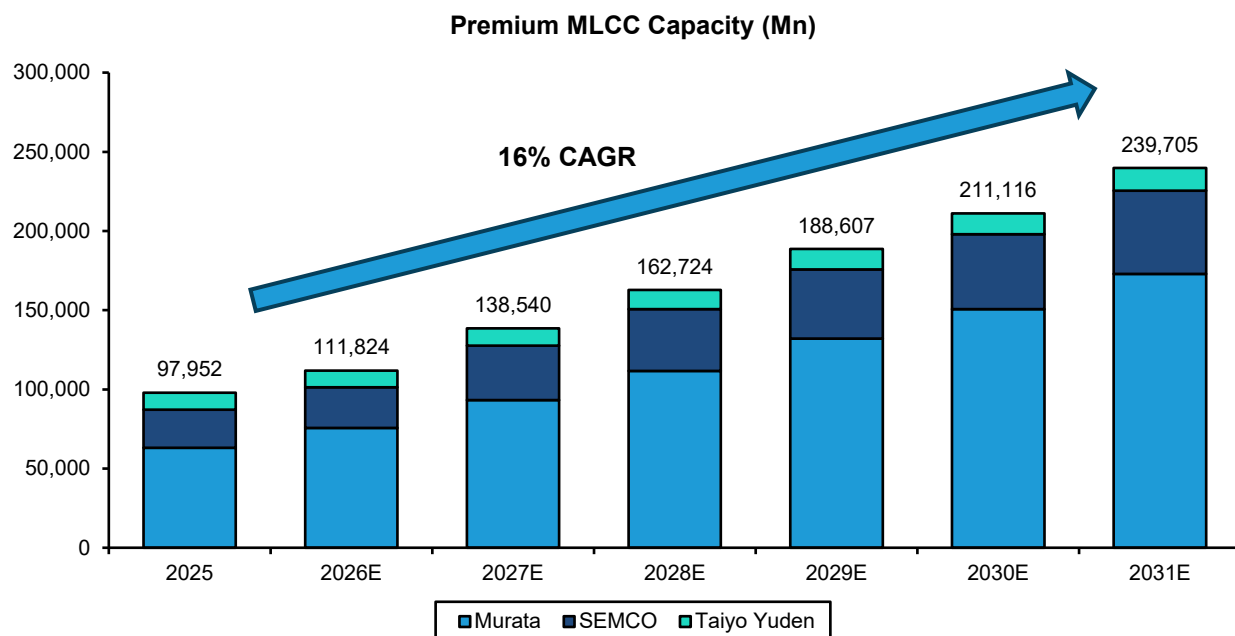
EXHIBIT 33: Miniaturization- Process of creating high end MLCC requires differentiated stacking, printing, packaging and firing offers processes

Multilayer Ceramic Capacitor Manufacturing Process



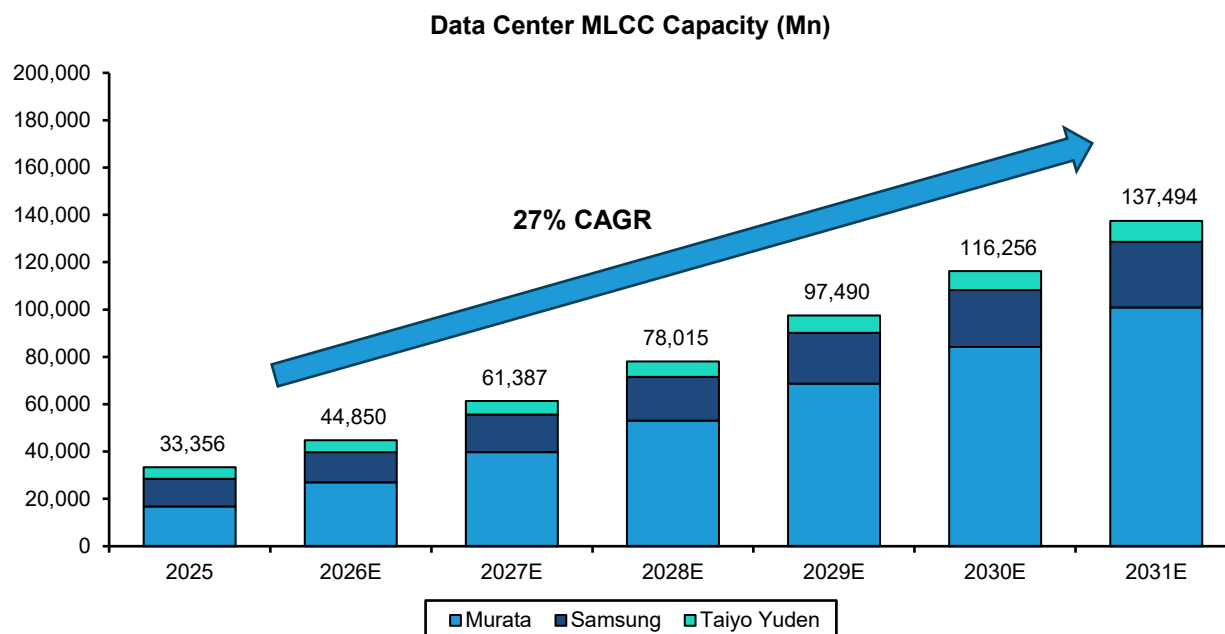
Source: Bernstein analysis

EXHIBIT 34: The combined premium MLCC capacity of Murata, SEMCO, and Taiyo Yuden is expected to grow at a 16% CAGR



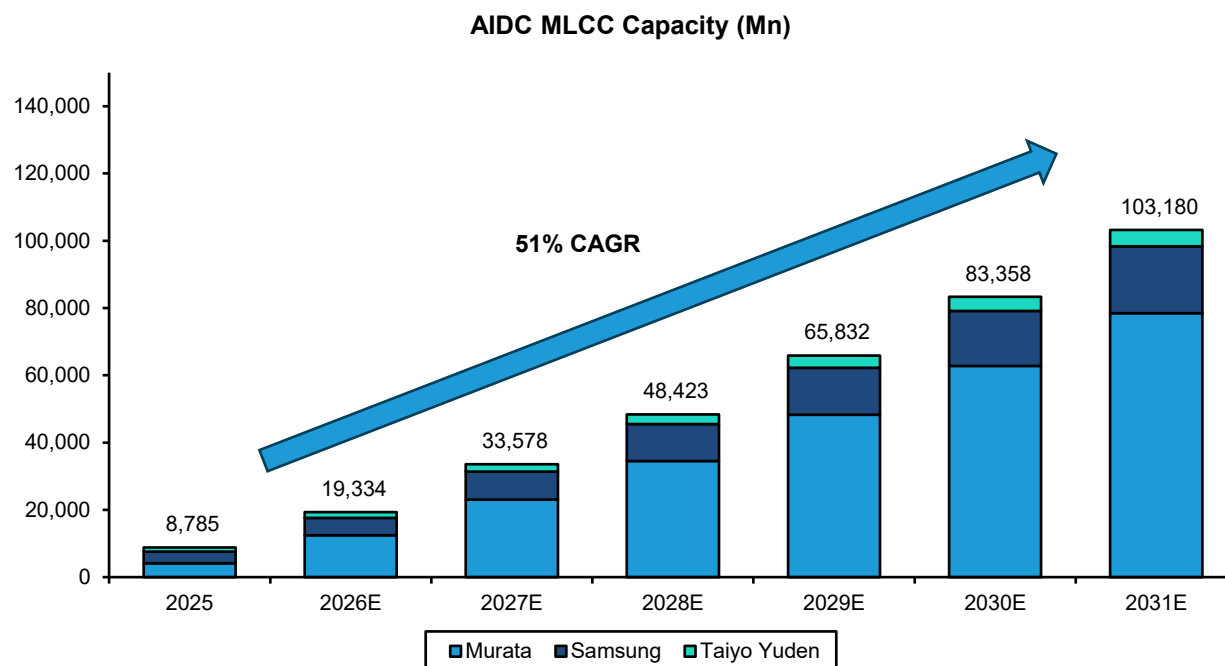
Premium includes those for Data center and automotive
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 35: Premium MLCC players are increasingly focused on data-center capacity, which is expected to grow at a 27% CAGR to 137bn units by 2031E

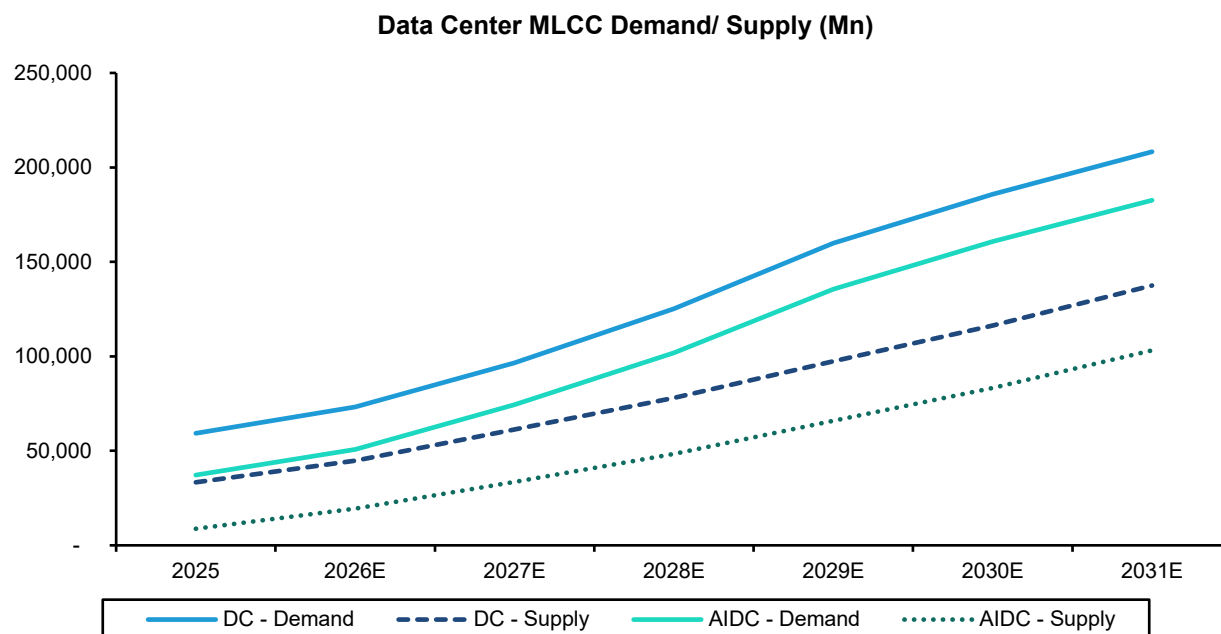


Source: Company reports, Bernstein estimates and analysis

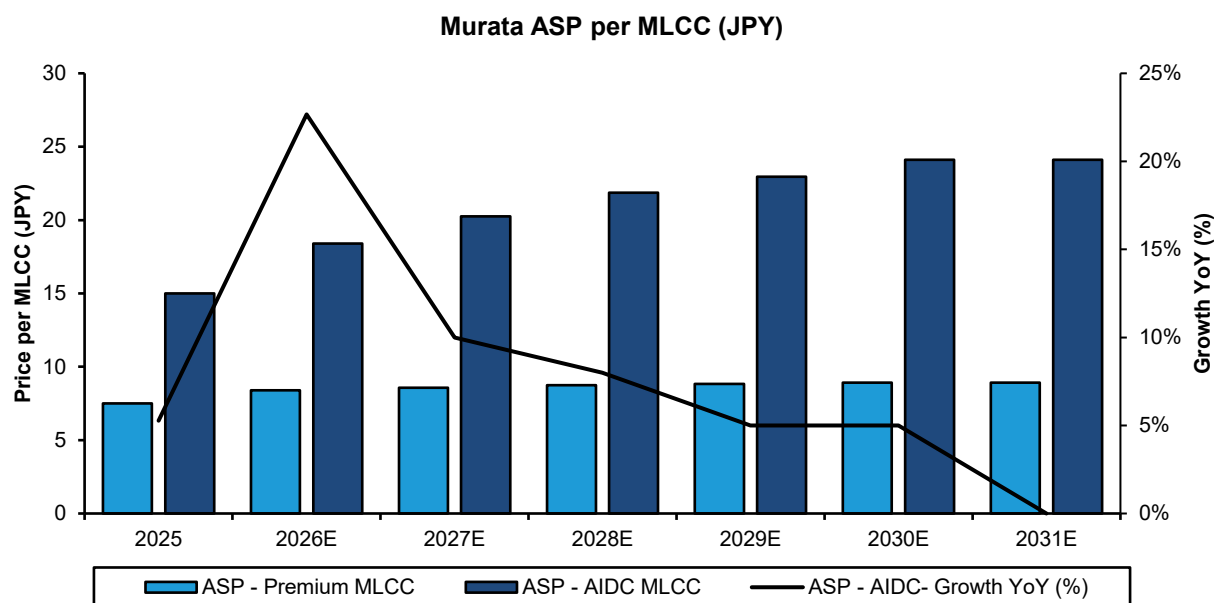
EXHIBIT 36: Murata to outpace peers in AI data-center MLCCs on superior technology and production know-how, with overall market capacity growing at a 51% CAGR through 2031E



Source: Company reports, Bernstein estimates and analysis

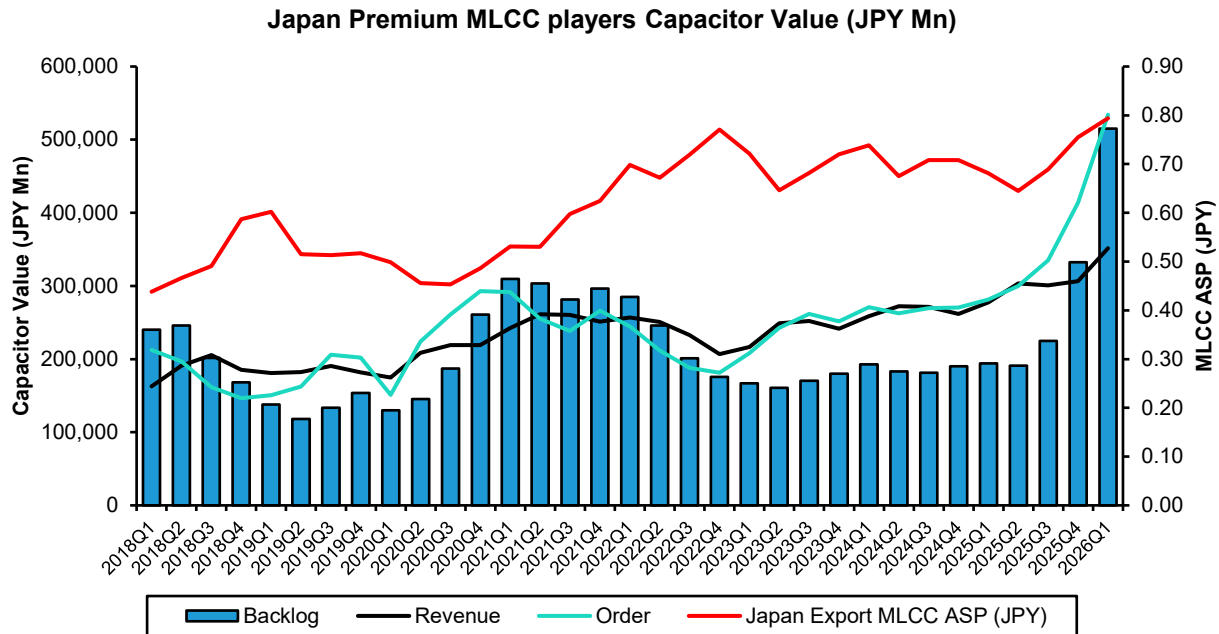
EXHIBIT 37: AI data center MLCC supply to remain tight despite capacity expansion, supporting further price upside

Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 38: Cutting-edge MLCC prices are expected to rise 15-30% in the near term, followed by 5-10% longer-term growth amid tight supply-demand conditions

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 39: AI DC MLCC demand is spilling over into the broader MLCC market, tightening industry capacity and keeping prices elevated for an extended period, as seen in previous supply cycles



Japan premium MLCC players refer to Murata, Taiyo Yuden (Not covered)
Source: Company disclosure, Ministry of Finance Japan, Bernstein analysis

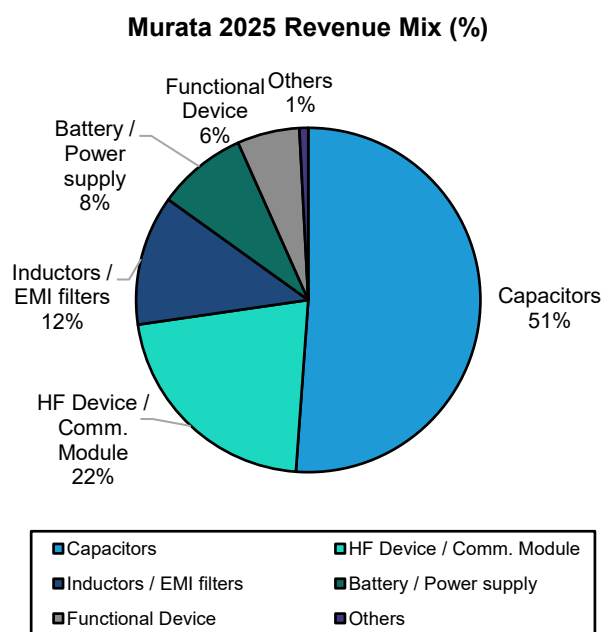
In our view, Murata is the key beneficiary from AI Data Center demand. Murata's technological edge in miniaturization and ceramic manufacturing enables it to produce high-end and cutting-edge MLCCs (47-100 microfarad with the standard size of 0402-0603) effectively at lower costs. We estimate DC takes up almost 20% of Murata's Capacitor revenue while Autos take up almost 35%. SEMCO is weaker on high-end and even weaker on cutting-edge technology with a strategy to expand into both automotive and DC MLCCs. As such, we expect Murata to grab share in AIDC MLCC from 47% in 2025 to 76% in 2031E. Murata's MLCC mix shift and higher ASP from cutting-edge MLCC would drive total capacitor revenue up by +23% CAGR in the next 5 years.

- Capacitors take up 51% of Murata's revenue and Data Center takes up ~9% as a portion of the end use market.** MLCCs take up almost all of Murata's capacitor's revenue with communication accounting for 36%, mobility for 26%, and computers, which includes datacenter, accounting for 17%. Historically, high-end communication applications, such as smartphones, were Murata's core market. However, the revenue mix is gradually shifting toward mobility and computing as premium MLCC demand continues to expand across these applications. Within Murata's capacitor business, we estimate that communications accounted for 25% of revenue in 2025, compared with 33% for mobility and 17% for data centers, highlighting the growing importance of premium MLCC applications. (*Exhibit 40 - Exhibit 43*)
- Murata's technological edge in advanced premium MLCC gives the company a head start with additional capacity expansion.** Murata's current capability to produce cutting edge MLCCs for AI data center means it can scale capacity even faster than its peers. This is through manufacturing and supply chain expansion know-how that peers are unfamiliar with given their initial entry to the AI cutting edge market today. Murata has committed another JPY40Bn annually for the next 2 years to expand their AI cutting edge MLCC factories in addition to their 10-15% normal annual capacity expansion program. While it is unclear that the 10-15% capacity expansion refers to which set of lines or application or how uniform is this expansion, we expect this applies to the broader premium segment (Computers, Data Centers, Automotive) with AI DC experiencing a much larger increase within this segment. (*Exhibit 44, Exhibit 45*)
- Murata focuses on AI DC while closest peer SEMCO focuses on automotive and AI DC.** Among the two leaders that can produce cutting edge MLCCs at scale, Murata's technological edge in miniaturization and ceramic manufacturing enables it to produce high-end and cutting-edge MLCCs effectively at lower costs. SEMCO is relatively weaker on high-end and even weaker on cutting-edge technology. SEMCO's most recent strategy comments in the last 2 quarters signal their intention to expand into both automotive and DC MLCCs in the range of 20% capacity expansion. We question the

effectiveness of this capacity expansion especially in cutting edge segment as channel checks reveal SEMCO's yield is not as mature resulting in higher prices or lower margins.

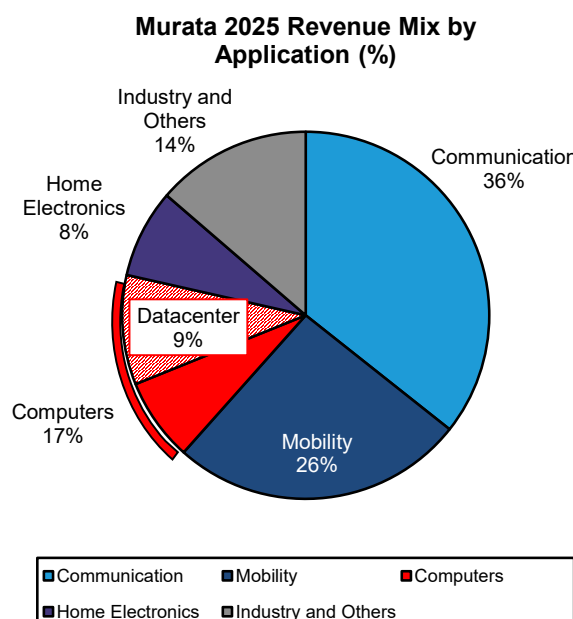
- Recent end of life notice for generic volume based and lower end autos MLCCs signals Murata's focus to continue moving up the technology curve.** In early September, many news outlets have reported Murata issuing end of life notice for some MLCCs in consumer, industrial, networking, telecommunication, automotive infotainment and comfort related applications. Major automotive MLCCs are not discontinued including those related to battery rails, ADAS, and powertrains that require higher capacitance and harsher conditions. AI DC MLCCs require more layers, tighter process control, and consume more effective production capacity than commodity products, making the phase-out of lower-end products an important step in freeing up capacity for cutting-edge MLCCs. This shift strategy should help Murata reduce its exposure to cyclical, supply-demand-driven commodity markets with numerous competitors and shift toward more differentiated, higher-value products. As AI data center demand continues to grow, we expect Murata to accelerate this shift, leveraging its technological lead and differentiated position relative to competitors.
- As such, this means that Murata can sufficiently capture market share in premium AI DC space.** As data center specifications and orders are usually quite demanding in terms of order volume and lead time, Murata's ability to produce more, faster and at higher quality means they can qualify as the main provider on reference boards (usually 70% share). The customer stickiness moat then kicks in as mission critical data center vendors prefer the same vendor for reliability and scale at procurement. We expect Murata's share in AIDC MLCC to increase from an estimated 47% to around 76% in 2031E. Murata's premium MLCC revenue would then grow at 33% CAGR and Capacitor revenue by 23% CAGR till 2031E predominantly on mix shift towards higher end and more expensive MLCCs.

EXHIBIT 40: **Murata's revenue is diversified across Components, Modules/Devices, and Sensors, led by MLCC**



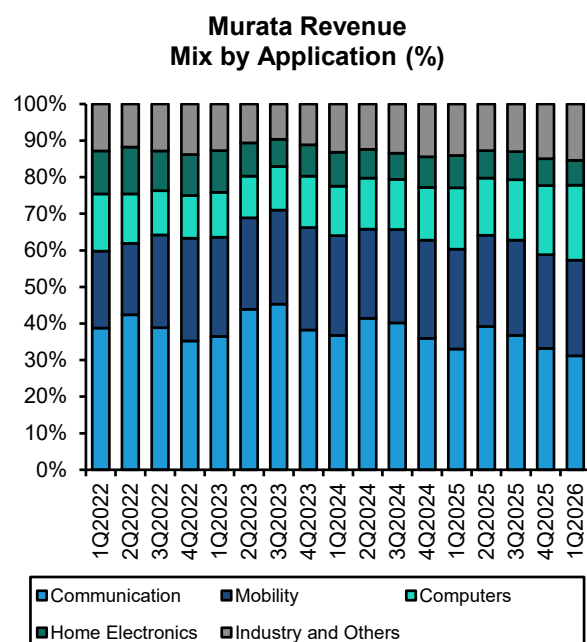
Source: Company reports, Bernstein analysis

EXHIBIT 41: **Murata's revenue proportion to the Greater China region has declined overtime, depicting localization trend**



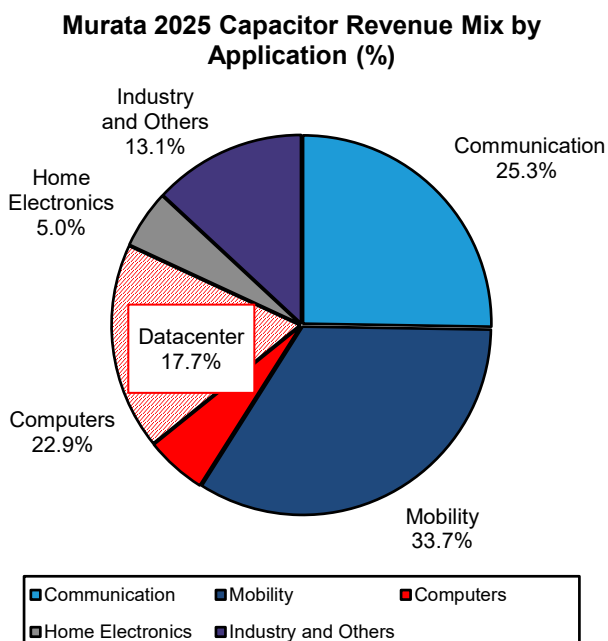
Source: Company reports, Bernstein analysis

EXHIBIT 42: **Murata's revenue mix is shifting from communications toward premium mobility and data-center applications**



Source: Company reports, Bernstein analysis

EXHIBIT 43: **Murata's capacitor revenue is concentrated in premium mobility and computing applications**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 44: **Murata solidifies leadership across key growth segments with its technology excellence**

	Murata	Samsung Electro	TDK	Taiyo Yuden	Yageo	Chinese (Fenghua, etc.)
Global MLCC Market share	40%	23%	8~12%	7~10%	7~10%	>10%
Smallest MLCC size	006003 size	008004 size	008004 size	008004 size	01005 size	01005+ size
High-cap Flagship MLCC (AI DC)	0402 size 47μF / 2.5V	0402 size 47μF / 2.5V	0402 size 1μF / 4V	0402 size 4.3μF / 2.5V	0402 size 0.1μF / 4V	0402 size 2.2μF / 6.3V
High-Voltage C0G grade (EV OBC)	1210 size 15nF / 1.25kV	1210 size 10nF / 1.25kV	1210 size 10nF / 1.25kV	1210 size 22nF / 1kV	1210 size 44nF / 1kV	1210 size 0.33nF / 1kV
ADAS Capabilities	1210 size 220μF / 4V	1210 size 100μF / 4V	1210 size 100μF / 4V	1210 size 220μF / 4V (Derating)	N/A (Automotive grade x)	N/A (Automotive grade x)
Space Capabilities	Yes (JAXA certificate)	Rumor (26.1Q call)	N/A	N/A	N/A	N/A

Areas where Murata has competitive advantage versus peers.
 Areas where peers are not active

Samsung Electro-mechanics, TDK, Taiyo Yuden, Yageo, and Fenghua are not covered by Bernstein

Source: Company disclosure, Bernstein analysis

EXHIBIT 45: Murata is expanding its high-end capacity of 10-15%YoY to meet high premium MLCC demand and solidify its market share in Data Center

Company	Region	Key Product	Average 21-25. CAPEX (JPY Bn)	26F CAPEX (JPY Bn)	Detail
Yasu plant	Japan	Ceramic Materials	3	15	• R&D for materials and production technology
Fukui Manufacturing ^(*)	Japan	MLCC – Cutting edge	29	32	• Comprehensive R&D center for MLCC prototypes and new product development
Izumo Manufacturing ^(*)	Japan	MLCC – Cutting edge	14	58	• Production building completed in April 2026, largest scale in Japan; capacity expansion planned through future equipment installation
Wuxi Murata	China	MLCC – General	22	12	• Additional expansion completed in 2025, largest overseas facility
Philippine Murata	Philippine	MLCC – General, Vehicle	16	8	• Focused general-purpose MLCC production; new production building completed in 2025; capacity expansion expected focusing on vehicle applications
Murata Electronics (Thailand)	Thailand	MLCC – Vehicle, Communication	8	20	• Production building completed in 2024; Additional expansion planned through 2028
Total			92	145	• Plans to execute additional investments JPY80Bn over 2 years for capacity expansion

(*) Including facilities operated by subsidiaries

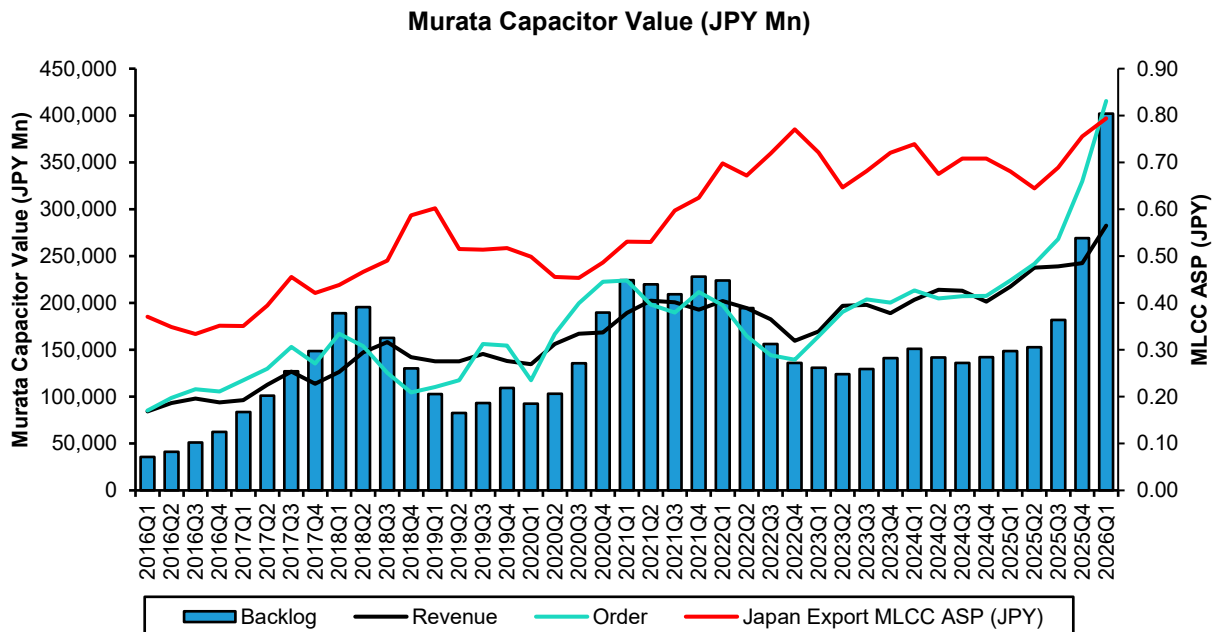
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 46: Premium players are increasing capex to capture surging demand for high-end MLCCs, while Murata further extends its lead in cutting-edge MLCCs

Company	Key Strategy	Market Share	Utilization	CAPEX Forecast (2026E)	High-End Capacity Expansion (%)	Detail
Murata	Leader in MLCC, Further Expanding Cutting-edge MLCC Production	40% (High-end 55%)	>95%	JPY250Bn (Additional capex JPY 40Bn for MLCC)	~20% Annually	- Completed construction of the Izumo Plant, the largest domestic facility. The plant will lead future capacity expansion for high-end MLCCs. Recently announced EOLs for general-purpose products, further accelerating the transition to Cutting-edge products - Planned JPY 80Bn in additional production investment over the next two years, implying annual capacity growth ~10%, plus 10~15% additional capacity expansion
Samsung Electro-Mechanics	Pivoting product mix to High-end for AI DC and Autos	23% (High-end 25%)	95%	Total KRW 3.0Tn (USD 2.2Bn) KRW1.0Tn (USD 0.8Bn) for MLCC	~30% from 2H27 (10% Annually)	- Shifted product mix toward premium, higher-end products, as reflected in ASP improvement despite capacity declining from 2024; capacity expansion beginning in 1Q26 - Philippines plant construction to begin in 2026 for AI-focused premium products, with production starting in 2H27 - Secured long-term supply agreements with Hyperscalers, accelerating the move to high-end MLCCs.
Taiyo Yuden	Enhancing High-end capabilities to capture emerging opportunities alongside Murata/SEMCO	10% (High-end 10%)	90~95%	JPY40Bn	~10% Annually	- Large-scale facility build-out to be completed by 2025, with capacity expansion expected from 2026 onward (JPY 270Bn over 5yrs) - Targeting growth opportunities in MLCC and inductors driven by AI server and EV demand - Capacity expected to expand at ~10% CAGR on an annual basis (targeting 15% in FY27)

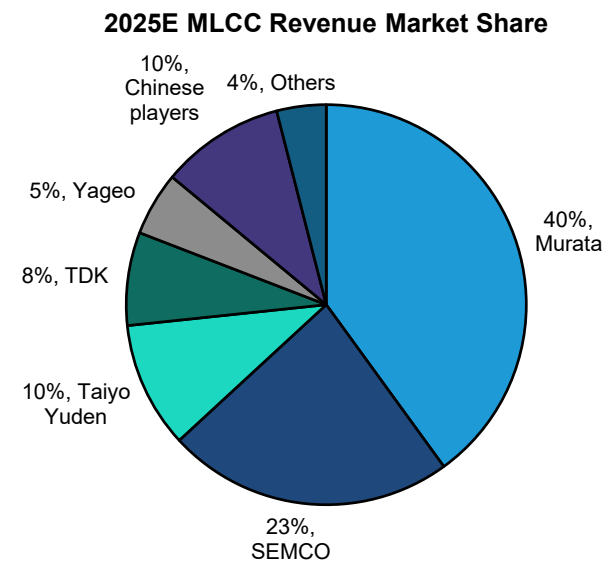
Source: Company disclosures, Bernstein analysis

EXHIBIT 47: We expect prices to continue staying high as MLCC is facing capacity constraints from pull forward demand historically lasting 18-24 months



Source: Company disclosure, Ministry of Finance Japan, Bernstein analysis

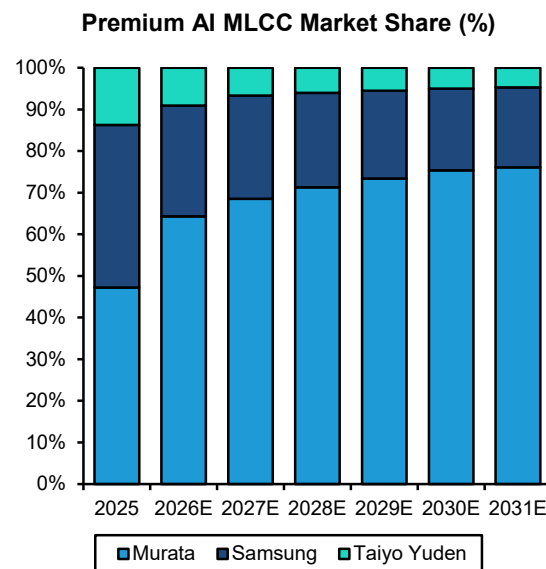
EXHIBIT 48: Murata has 40% market share in total MLCC



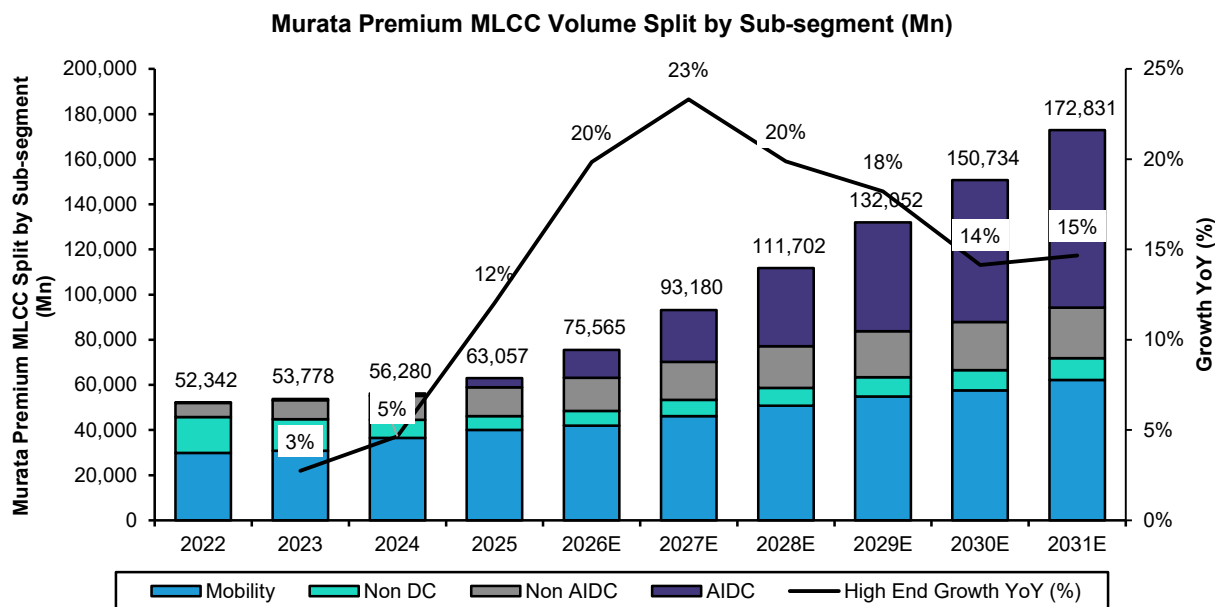
Samsung Electro-Mechanics, TDK, Taiyo Yuden, Yageo, and Viyong are not covered by Bernstein

Source: Bloomberg, Bernstein estimates and analysis

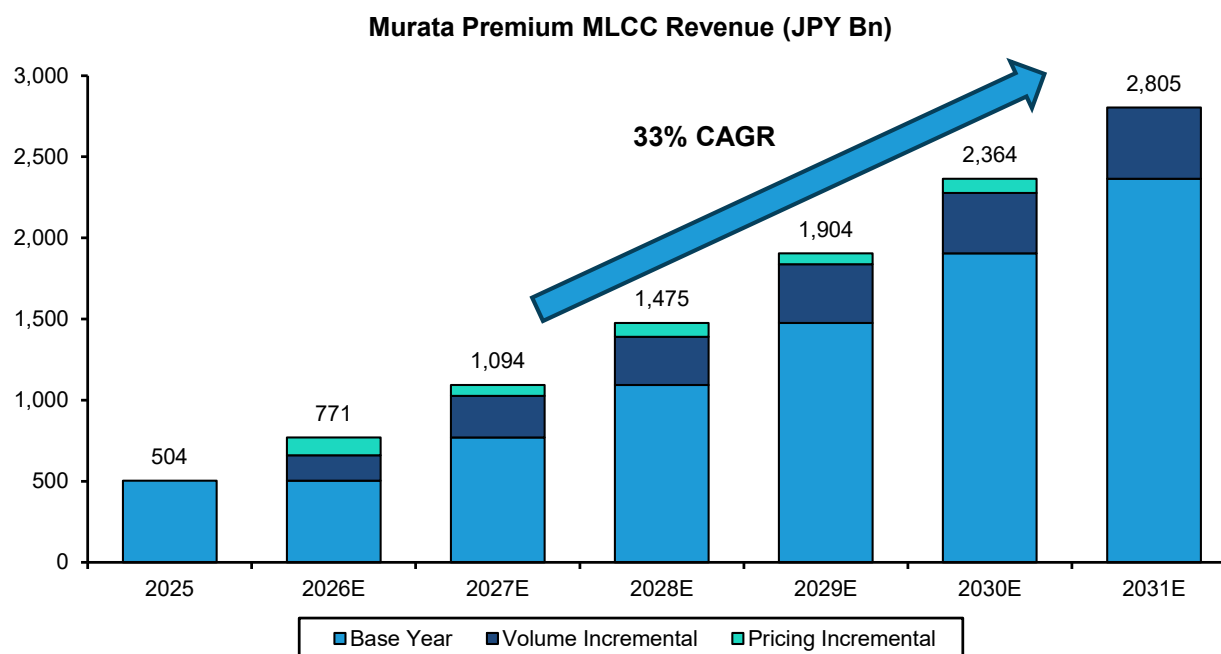
EXHIBIT 49: Murata should capture disproportionate AI upside because premium MLCC mix, share, and capacity conversion all moves in its favor



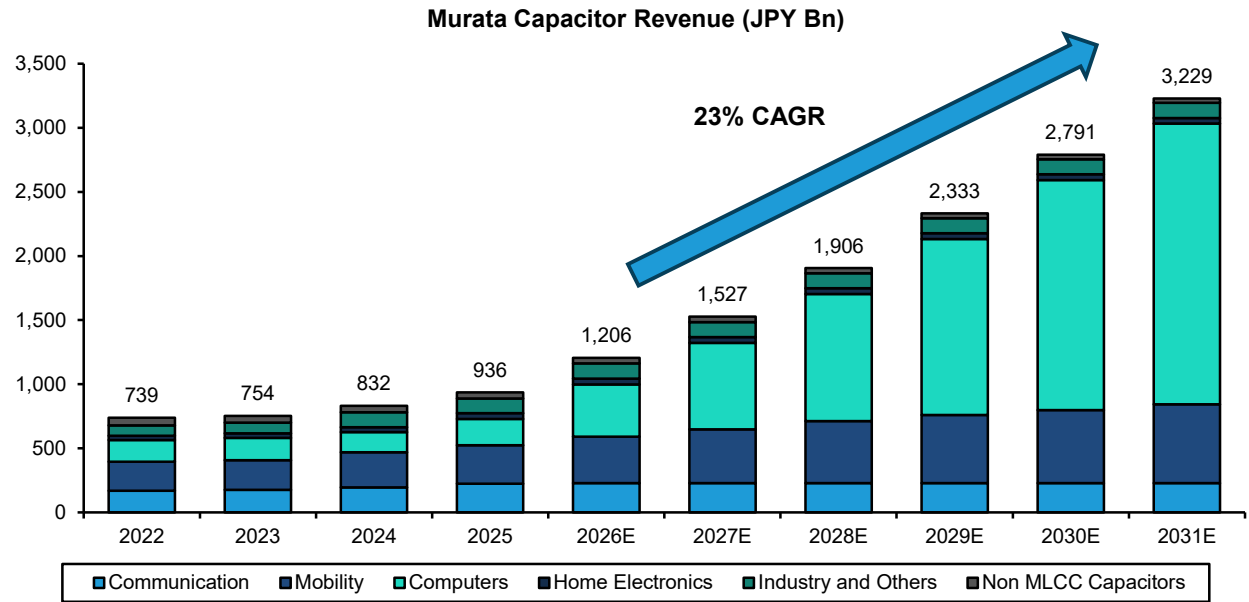
Samsung Electro-mechanics and Taiyo Yuden are not covered by Bernstein
 Source: Bloomberg, Bernstein estimates and analysis

EXHIBIT 50: **We expect Murata to allocate a larger volume mix of its premium MLCC to AIDC/DC**

Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 51: **Murata's reluctance to increase prices means premium MLCC's growth is restricted to mainly volume capacity growth**

Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 52: **We expect Murata's overall Capacitor Revenue to grow by +23%YoY CAGR driven by AI Data Center**

Source: Company disclosures, Bernstein estimates and analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 53: Murata Income Statement

Income Statement

JPY billion	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Revenue	1,136	1,372	1,575	1,534	1,630	1,813	1,687	1,640	1,743	1,831	2,187	2,666	3,188	3,701	4,214	4,738
Gross profit	432	445	600	581	622	768	674	637	718	775	977	1,257	1,590	1,904	2,246	2,592
Selling, general and administrative expenses	(149)	(188)	(232)	(230)	(204)	(233)	(256)	(243)	(278)	(297)	(357)	(386)	(401)	(414)	(434)	(434)
Research and development expenses	(82)	(94)	(102)	(102)	(102)	(111)	(125)	(133)	(149)	(159)	(188)	(224)	(266)	(305)	(344)	(382)
Net other income (expense)	0	0	0	0	0	0	5	(46)	(11)	(37)	5	6	6	6	6	6
Operating Profit	201	163	267	253	313	424	298	215	280	282	474	682	945	1,204	1,494	1,781
EBITDA	315	305	391	394	456	580	468	391	453	460	657	868	1,141	1,415	1,726	2,039
Net other financial income (expense)	2	2	3	4	1	1	4	24	25	27	25	28	43	65	95	134
Share of profits (losses) of investments accounted for using th	(5)	(7)	(2)	(4)	(10)	0	0	0	0	0	0	0	0	0	0	0
Others	2	3	0	1	12	8	0	0	0	0	0	0	0	0	0	0
Profit before tax	200	168	267	254	316	433	303	239	304	309	498	710	988	1,269	1,589	1,916
Income tax expense	(44)	(22)	(60)	(71)	(79)	(119)	(59)	(59)	(71)	(75)	(125)	(177)	(247)	(317)	(397)	(479)
Net income	156	146	207	183	237	314	244	180	233	234	373	532	741	951	1,192	1,437
Basic EPS (JPY)	81.5	73.2	107.8	95.4	123.5	163.7	128.6	95.7	125.1	127.7	205.1	292.6	407.1	522.9	654.9	789.7
Margins																
Gross profit	38.1%	32.4%	38.1%	37.9%	38.2%	42.4%	39.9%	38.8%	41.2%	42.3%	44.7%	47.2%	49.9%	51.4%	53.3%	54.7%
Operating Profit	17.7%	11.9%	16.9%	16.5%	19.2%	23.4%	17.7%	13.1%	16.0%	15.4%	21.7%	25.6%	29.6%	32.5%	35.4%	37.6%
Profit before tax	17.6%	12.2%	17.0%	16.6%	19.4%	23.9%	17.9%	14.6%	17.5%	16.9%	22.8%	26.6%	31.0%	34.3%	37.7%	40.4%
Net income	13.7%	10.6%	13.1%	11.9%	14.5%	17.3%	14.4%	11.0%	13.4%	12.8%	17.1%	20.0%	23.2%	25.7%	28.3%	30.3%
Growth YoY																
Revenue		20.8%	14.8%	-2.6%	6.3%	11.2%	-6.9%	-2.8%	6.3%	5.0%	19.4%	21.9%	19.6%	16.1%	13.9%	12.4%
Gross profit		2.9%	34.9%	-3.1%	7.0%	23.5%	-12.3%	-5.5%	12.7%	8.0%	26.1%	28.7%	26.5%	19.7%	18.0%	15.4%
Operating Profit		-18.9%	63.4%	-5.1%	23.7%	35.4%	-29.7%	-27.8%	29.8%	0.8%	68.1%	44.0%	38.5%	27.4%	24.1%	19.3%
Profit before tax		-16.3%	59.3%	-5.0%	24.6%	36.8%	-30.0%	-20.9%	27.2%	1.4%	61.5%	42.4%	39.1%	28.4%	25.2%	20.6%
Net income		-6.4%	41.7%	-11.6%	29.5%	32.4%	-22.4%	-26.0%	29.2%	0.3%	59.7%	42.7%	39.1%	28.4%	25.2%	20.6%
EPS		-10.2%	47.3%	-11.6%	29.5%	32.5%	-21.4%	-25.6%	30.7%	2.1%	60.7%	42.6%	39.1%	28.4%	25.2%	20.6%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 54: Murata Balance Sheet

Balance Sheet

JPY billion	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Cash and cash equivalents	344	200	252	347	428	544	469	622	625	654	747	1,059	1,533	2,208	3,084	4,205
Trade receivables	209	258	274	281	341	358	270	293	294	328	388	456	529	591	657	709
Inventories	211	290	349	334	361	465	575	513	483	520	609	681	754	819	878	923
Other financial assets	53	21	27	30	23	24	43	20	51	24	43	43	43	43	43	43
Other current assets	28	42	31	36	31	44	50	52	45	55	86	86	86	86	86	86
Total current assets	846	810	934	1,027	1,184	1,435	1,408	1,500	1,498	1,581	1,872	2,325	2,944	3,746	4,748	5,966
Property, plant and equipment	508	705	856	1,002	1,079	1,085	1,146	1,185	1,184	1,301	1,356	1,455	1,593	1,756	1,958	2,187
Right-of-use assets	0	0	0	0	32	32	50	59	64	60	60	60	60	60	60	60
Goodwill	62	77	78	73	71	118	123	137	136	99	96	91	85	80	76	71
Intangible assets	49	54	48	39	34	47	39	39	41	36	25	17	14	13	14	15
Other financial assets	103	87	67	49	41	35	39	47	40	50	59	59	59	59	59	59
Deferred tax assets	5	36	42	42	38	27	43	47	53	60	64	64	64	64	64	64
Other non-current assets	36	27	24	18	15	31	10	23	13	10	11	11	11	11	11	11
Total Non-current assets	764	987	1,115	1,223	1,310	1,374	1,451	1,538	1,530	1,618	1,671	1,757	1,887	2,044	2,241	2,466
Total Assets	1,609	1,797	2,049	2,250	2,494	2,809	2,858	3,038	3,028	3,199	3,543	4,082	4,831	5,790	6,989	8,432
Bonds and borrowings	0	0	0	0	40	0	61	50	1	2	1	1	1	1	1	1
Trade payables	58	94	75	79	86	94	66	68	70	80	85	97	111	124	137	149
Lease liabilities	0	0	0	7	7	6	8	9	10	11	11	11	11	11	11	11
Other financial liabilities	0	0	0	0	53	64	60	56	49	57	57	57	57	57	57	57
Income taxes payable	7	25	38	28	42	59	13	27	31	43	27	27	27	27	27	27
Deferred income	0	0	0	0	0	0	1	1	1	1	2	2	2	2	2	2
Provisions	0	0	0	0	0	0	3	2	2	3	5	5	5	5	5	5
Other current liabilities	0	0	0	0	91	104	88	105	113	119	104	104	104	104	104	104
Total Current liabilities	209	262	260	284	318	321	298	318	277	316	291	304	318	331	344	356
Non-current liabilities																
Bonds and borrowings	1	1	100	150	111	111	51	2	2	2	2	2	2	2	2	2
Lease liabilities	0	0	0	0	0	0	0	0	0	0	0	0	42	42	42	42
Other financial liabilities	0	0	0	0	0	0	3	4	5	1	1	1	1	1	1	1
Deferred income	0	0	0	0	0	0	19	20	19	22	21	21	21	21	21	21
Retirement benefit liability	59	68	76	85	74	65	70	71	75	72	73	73	73	73	73	73
Provisions	0	0	0	0	0	0	9	8	7	10	9	9	9	9	9	9
Deferred tax liabilities	10	8	7	6	8	13	9	11	12	10	10	10	10	10	10	10
Other non-current liabilities	1	2	2	2	5	10	3	4	4	6	7	7	7	7	7	7
Total Non-current liabilities	71	78	185	271	223	224	200	165	172	165	165	165	165	165	165	165
Total Liabilities	280	340	444	555	541	545	498	482	448	481	457	469	483	496	509	521
Murata Manufacturing Co., Ltd. stockholders' equity	1,355	1,457	1,604	1,694	1,921	2,264	2,360	2,556	2,581	2,719	3,088	3,614	4,350	5,295	6,481	7,913
Non-controlling interests	1	1	1	1	1	0	0	-1	-1	-1	-1	-1	-1	-1	-1	-1
Total Equity	1,355	1,457	1,605	1,695	1,922	2,264	2,360	2,556	2,580	2,718	3,087	3,613	4,349	5,294	6,480	7,912
Total Liabilities and Equity	1,635	1,797	2,049	2,250	2,462	2,809	2,858	3,038	3,028	3,199	3,544	4,083	4,831	5,790	6,989	8,432

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 55: **Murata Cash Flow Statement**

Cash Flow Statement

JPY billion	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Net Income	156	146	207	183	237	314	244	180	233	234	373	532	741	951	1,192	1,437
Depreciation & amortization	114	142	124	140	143	156	170	176	173	178	184	186	196	211	233	258
Changes in working capital	(14)	(65)	(90)	(5)	(50)	(76)	(42)	83	29	(23)	(166)	(128)	(132)	(114)	(112)	(85)
Other	5	(21)	24	41	4	1	56	74	69	96	104	150	204	252	302	345
Cash flow from operations	0	0	0	0	0	0	0	0	(7)	(7)	(7)	(7)	(7)	(7)	(7)	(7)
Capex	(218)	(320)	(317)	(313)	(226)	(189)	(228)	(241)	(197)	(257)	(232)	(272)	(325)	(369)	(430)	(483)
Net proceeds from investment income	45	57	29	31	59	30	54	30	27	23	22	32	47	70	100	139
Other	(36)	(30)	1	9	10	8	24	9	(38)	41	(19)	0	0	0	0	0
Cash flow from investments	(203)	(194)	(304)	(284)	(150)	(212)	(151)	(202)	(208)	(194)	(79)	(239)	(278)	(299)	(330)	(344)
Cash flow from financing	(12)	(84)	52	18	(118)	(118)	(182)	(165)	(243)	(222)	(34)	(11)	(10)	(10)	(10)	(10)
Net effect of exchange rate changes	(3)	1	2	1	0	13	13	30	2	19	4	0	0	0	0	0
Net change in cash	27	(51)	30	85	105	104	(43)	153	3	29	93	312	474	675	877	1,121
Beginning cash	663	977	899	1,046	1,249	1,806	1,867	1,947	2,305	2,309	2,406	3,241	4,744	6,971	10,004	13,929
Ending cash	902	926	929	1,131	1,354	1,911	1,824	2,100	2,308	2,338	2,499	3,553	5,217	7,646	10,880	16,050

Source: Company reports, Bernstein estimates and analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Murata Manufacturing Co Ltd**

We rate Murata Outperform, with target price of ¥10,000. We used an P/E valuation, applying 30.0x P/E on 2 Year Blended Forward EPS and triangulate using a DCF (WACC: 8%, PGR: 3%).

RISKS**Murata Manufacturing Co Ltd**

The downside risks include: 1) Slowdown of MLCC demand from the slowdown in Data Center related demand. 2) Capacity expansion cannot keep up with demand, resulting in lower than expected revenue growth. 3) Peer's technology catch up among high end MLCC increasing competition and resulting in a price war.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

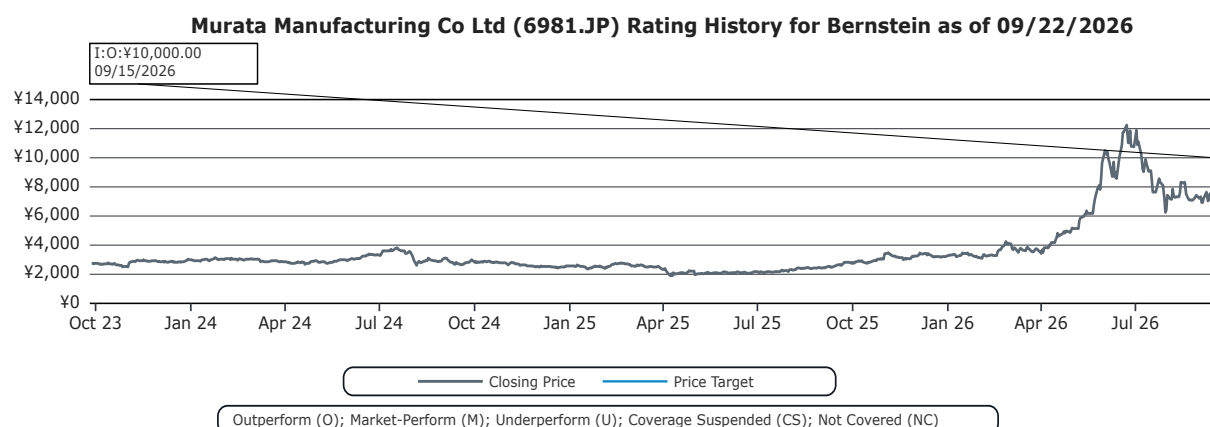
It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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